

MTH-RA, Real Analysis

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General Information About the Course

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The classical motivation for measure theory goes back more than two thousand years: the Greeks wanted to measure the area of a circle, the volume of a ball, and other regions in the plane and in space. This eventually evolved into an abstract theory: instead of working only in $\mathbb{R}^d$, we work with an arbitrary set Ω and ask how to assign a consistent notion of “size” to certain of its subsets.

A key obstruction appears immediately:

- It is **not possible**, in general, to consistently measure *every* subset of a set (even of $\mathbb{R}^d$).
- Instead, we single out a suitable collection of “good” subsets (we call them the **measurable sets**) and only attempt to measure those.

This leads to the first fundamental definition of the course: the notion of a σ -algebra.

Definition 1.1 (σ -algebra). Let Ω be a set. A collection Σ of subsets of Ω is called a σ -algebra (or σ -field) on Ω if:

- (i) $\Sigma \neq \emptyset$ (equivalently, $\Omega \in \Sigma$);
- (ii) (closed under complements) if $A \in \Sigma$, then $A^c := \Omega \setminus A \in \Sigma$;
- (iii) (closed under countable unions) if $A_1, A_2, A_3, \dots \in \Sigma$, then $\bigcup_{n=1}^{\infty} A_n \in \Sigma$.

The pair (Ω, Σ) is called a **measurable space**, and the sets $A \in \Sigma$ are called **measurable sets**.

Remark 1.1. Condition (i) simply rules out the degenerate empty collection, which would vacuously satisfy (ii) and (iii) but is of no interest. Some textbooks instead require $\emptyset \in \Sigma$ directly; this is equivalent given (ii)–(iii) and a nonempty Σ .

Proposition 1.1

Every σ -algebra Σ on Ω satisfies:

- (a) $\emptyset \in \Sigma$ and $\Omega \in \Sigma$;
- (b) Σ is closed under countable intersections: if $A_1, A_2, \dots \in \Sigma$ then $\bigcap_{n=1}^{\infty} A_n \in \Sigma$;
- (c) Σ is closed under finite unions and finite intersections (in particular under set differences).

Proof. (a) Since $\Sigma \neq \emptyset$, pick $A \in \Sigma$. By (ii), $A^c \in \Sigma$, and by (iii) (a union of the two, or of A and infinitely many copies of it), $A \cup A^c = \Omega \in \Sigma$. Applying (ii) again, $\Omega^c = \emptyset \in \Sigma$.

(b) By De Morgan's law,

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c \right)^c. \quad (1.1)$$

Each $A_n^c \in \Sigma$ by (ii), the union is in Σ by (iii), and its complement is in Σ by (ii) again.

(c) A finite union $A_1 \cup \dots \cup A_k$ is a countable union with $A_n = \emptyset$ for $n > k$ (using $\emptyset \in \Sigma$ from (a)), so it lies in Σ by (iii). Finite intersections follow similarly from (b), and $A \setminus B = A \cap B^c \in \Sigma$. ■

Remark 1.2. The definition does *not* require the sets A_n in condition (iii) to be pairwise disjoint; the countable union of *any* sequence of measurable sets is measurable.

Example 1.1 (Trivial and power set σ -algebras)

For any Ω :

- $\{\emptyset, \Omega\}$ is the smallest possible σ -algebra on Ω .
- The power set $\mathcal{P}(\Omega)$ (all subsets of Ω) is the largest possible σ -algebra on Ω ; if $|\Omega| = n < \infty$ then $|\mathcal{P}(\Omega)| = 2^n$.

Example 1.2 (Generated σ -algebra)

Let X be an arbitrary collection of subsets of Ω . There is a **smallest** σ -algebra $\sigma(X)$ containing X , called the σ -algebra **generated by** X . It can be constructed as

$$\sigma(X) = \bigcap \{ \Sigma : \Sigma \text{ is a } \sigma\text{-algebra on } \Omega, X \subseteq \Sigma \}, \quad (1.2)$$

i.e. the intersection of *all* σ -algebras containing X (an arbitrary intersection of σ -algebras is again a σ -algebra, and $\mathcal{P}(\Omega)$ is one such σ -algebra containing X , so this intersection is nonempty and well defined).

Lemma 1.2

Let Ω be a set and $\{\Sigma_\alpha\}_{\alpha \in J}$ an arbitrary collection of σ -algebras on Ω . Then $\bigcap_{\alpha \in J} \Sigma_\alpha$ is also a σ -algebra on Ω .

Proof. Every Σ_α contains $\emptyset$ and Ω , so $\Sigma := \bigcap_{\alpha \in J} \Sigma_\alpha$ is nonempty.

Suppose $A \in \Sigma$. Then $A \in \Sigma_\alpha$ for every α . Since Σ_α is a σ -algebra, it contains A^c , for every α . Therefore, $\Sigma := \bigcap_{\alpha \in J} \Sigma_\alpha$ also contains A^c .

Suppose $A_n \in \Sigma$ for $n \in \mathbb{N}$. Then $A_n \in \Sigma_\alpha$ for every α , for every n . Since Σ_α is a σ -algebra, it contains $\bigcup_{n \in \mathbb{N}} A_n$, for every α . Therefore, $\Sigma := \bigcap_{\alpha \in J} \Sigma_\alpha$ also contains $\bigcup_{n \in \mathbb{N}} A_n$. ■

Lemma 1.2 shows that $\sigma(X)$ from (1.2) is a σ -algebra.

Example 1.3 (A concrete generated σ -algebra)

Let $\Omega = \{1, 2, 3, 4\}$ and $X = \{\{1, 2\}, \{1, 4\}\}$. We compute $\sigma(X)$ by closing under complements and unions:

$$\{1, 2\}, \{3, 4\}, \{1, 4\}, \{2, 3\}, \{1, 2, 4\} \rightarrow \{1, 2\} \cup \{1, 4\}, \{3\}, \{2\}, \{1\}, \{4\}, \dots \quad (1.3)$$

Continuing this process eventually generates every subset of $\{1, 2, 3, 4\}$, so in this case $\sigma(X) = \mathcal{P}(\Omega)$, with 16 elements.

Example 1.4 (A proper subalgebra)

Let $\Omega = \{1, 2, 3, 4\}$ and $X = \{\{1, 2\}\}$. Closing under complement and union gives

$$\sigma(X) = \{\emptyset, \{1, 2\}, \{3, 4\}, \Omega\}, \quad (1.4)$$

which has only $4 = 2^2$ elements, which is a proper subset of $\mathcal{P}(\Omega)$ (which has 16 elements). This shows that a σ -algebra need not be the full power set.

Exercise 1.1 (Possible cardinalities). Let $\Omega = \{1, \dots, n\}$.

- (a) Show that every σ -algebra on a finite set Ω has cardinality equal to a power of 2.
- (b) For a fixed achievable cardinality 2^k , is the σ -algebra with that cardinality unique? Investigate how many distinct σ -algebras of cardinality 2^k exist on $\{1, \dots, n\}$.

Solution. Let Σ be a σ -algebra on Ω . We define a relation on Ω as follows:

$$x \sim y \iff \text{for every } A \in \Sigma, x \in A \text{ if and only if } y \in A. \quad (1.5)$$

This is clearly an equivalence relation. Hence, the equivalence classes partition Ω . Now, for a given $x \in \Omega$, let C_x be the equivalence class containing x . Then,

$$C_x = \bigcap_{A \in \Sigma, x \in A} A \in \Sigma, \quad (1.6)$$

as it's the intersection of sets in Σ . Furthermore, for $A \in \Sigma$,

$$A = \bigcup_{x \in A} C_x. \quad (1.7)$$

Hence, every element of Σ is the disjoint union of members of $\mathcal{C} = \{C_x : x \in \Omega\}$. Furthermore, **every** disjoint union of members of $\mathcal{C}$ is in Σ , since $\mathcal{C} \subseteq \Sigma$. Hence, the elements of Σ are in a bijection with the subsets of $\mathcal{C}$. Therefore,

$$|\Sigma| = 2^{|\mathcal{C}|}. \quad (1.8)$$

For the second part, $|\Sigma| = 2^k$ corresponds to a partition of Ω into k disjoint parts, which is the Stirling number of second kind, denoted as $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$. ■

Example 1.5 (Borel σ -algebra)

Let $\Omega = \mathbb{R}^d$ with its usual topology, and let X be the collection of all **open** subsets of $\mathbb{R}^d$. The generated σ -algebra

$$\mathcal{B}(\mathbb{R}^d) := \sigma(X) \quad (1.9)$$

is called the **Borel σ -algebra**. It contains every open set, every closed set (as complements of open sets), and all sets obtainable from these by countably many unions, intersections, and complements. In particular, on $\mathbb{R}$, it contains every interval (open, closed, or half-open).

It is useful to contrast a σ -algebra with a **topology**, since both are collections of subsets satisfying closure axioms.

Definition 1.2 (Topology). A **topology** τ on a set Ω is a collection of subsets of Ω (called **open sets**) such that:

- (i) $\emptyset, \Omega \in \tau$;
- (ii) τ is closed under **arbitrary** unions: if $\{O_j\}_{j \in J}$ is any family (not necessarily countable) of sets in τ , then $\bigcup_{j \in J} O_j \in \tau$;
- (iii) τ is closed under **finite** intersections: if $O_1, \dots, O_n \in \tau$, then $\bigcap_{i=1}^n O_i \in \tau$.

Remark 1.3 (Key differences). • A topology allows *arbitrary* (possibly uncountable) unions but only *finite* intersections.

- A σ -algebra allows only *countable* unions (and, as a consequence, countable intersections), but requires closure under **complements**, which a topology does not.

These are genuinely different axiom systems, even though both start from “a set together with a distinguished collection of subsets.”

Having fixed a measurable space (Ω, Σ) , we now assign a size to each measurable set.

Definition 1.3 (Measure). A **measure** on (Ω, Σ) is a function $\mu : \Sigma \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\} = [0, \infty]$ such that:

- (i) (countable additivity) if $A_1, A_2, A_3, \dots \in \Sigma$ are pairwise disjoint, then

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) = \sum_{n=1}^{\infty} \mu(A_n); \quad (1.10)$$

(ii) μ is not identically $+\infty$ (i.e. $\mu(A) < \infty$ for at least one $A \in \Sigma$).

The triple (Ω, Σ, μ) is called a **measure space**.

Remark 1.4 (Motivation for additivity). If a measure is meant to behave like a physical quantity, e.g. volume of water in a bucket, then combining two *disjoint* quantities should simply add their sizes: 3 liters and 5 liters combined give 8 liters, with nothing created or destroyed. Countable additivity is the natural axiomatization of this idea.

Proposition 1.3 (The measure of the empty set)

Let μ be a measure on (Ω, Σ) satisfying countable additivity (condition (i) above), and suppose μ is not identically $+\infty$. Then $\mu(\emptyset) = 0$.

Proof. Apply additivity to the disjoint sequence $A, \emptyset, \emptyset, \dots$ for a fixed A with $\mu(A) < \infty$:

$$\mu(A) = \mu(A \cup \emptyset) = \mu(A) + \mu(\emptyset). \quad (1.11)$$

Since $\mu(A) < \infty$, we may subtract it from both sides to get $\mu(\emptyset) = 0$.

(If instead $\mu(\emptyset) = c > 0$ were allowed, one could show $\mu(\emptyset) = \mu(\emptyset \cup \emptyset) = 2c$, forcing $c \in \{0, \infty\}$; and if $\mu(\emptyset) = \infty$ then, applying additivity with A and $\emptyset$, $\mu(A) = \mu(A) + \infty = \infty$ for every A , i.e. μ is identically ∞ . This is exactly the degenerate case excluded by (ii).) ■

Remark 1.5. Some textbooks instead simply impose $\mu(\emptyset) = 0$ directly as part of the definition, in place of the “not identically ∞ ” condition; the two conventions agree on all non-degenerate measures. This mirrors the analogous convention issue for topologies (where $\emptyset, \Omega \in \tau$ is imposed directly).

Definition 1.4. Let (Ω, Σ, μ) be a measure space.

- μ is a **finite measure** if $\mu(\Omega) < \infty$.
- If μ is finite and $\mu(\Omega) = 1$, then μ is called a **probability measure**.
- μ is **σ -finite** if there exists a sequence $A_1, A_2, A_3, \dots \in \Sigma$ such that

$$\mu(A_n) < \infty \text{ for every } n, \quad \text{and} \quad \Omega = \bigcup_{n=1}^{\infty} A_n. \quad (1.12)$$

Remark 1.6. If $\mu(\Omega) < \infty$, one can always **normalize** μ by setting $\tilde{\mu}(A) = \mu(A)/\mu(\Omega)$ (assuming $\mu(\Omega) > 0$), producing a probability measure with the same measurable sets of measure zero.

Proposition 1.4

Let (M, Σ, μ) be a measure space.

(a) If $A_1 \subseteq A_2 \subseteq \cdots$ is an increasing sequence of measurable sets, then

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) = \lim_{n \rightarrow \infty} \mu(A_n). \quad (1.13)$$

(b) If $A_1 \supseteq A_2 \supseteq \cdots$ is a decreasing sequence of measurable sets and $\mu(A_1) < \infty$, then

$$\mu \left(\bigcap_{n=1}^{\infty} A_n \right) = \lim_{n \rightarrow \infty} \mu(A_n). \quad (1.14)$$

Proof. (a) We define $B_1 = A_1$ and $B_n = A_n - A_{n-1}$ for $n \geq 2$. Then the B_n 's are pairwise disjoint, and $\bigcup_n B_n = \bigcup_n A_n$. As a result,

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) = \mu \left(\bigcup_{n=1}^{\infty} B_n \right) = \sum_{n=1}^{\infty} \mu(B_n) \quad (1.15)$$

$$= \lim_{N \rightarrow \infty} \sum_{n=1}^N \mu(B_n) = \lim_{N \rightarrow \infty} \mu \left(\bigcup_{n=1}^N B_n \right) \quad (1.16)$$

$$= \lim_{N \rightarrow \infty} \mu(A_N), \quad (1.17)$$

since $\bigcup_{n=1}^N B_n = A_N$.

(b) Define $B_n = A_1 - A_n$. Then B_n is an increasing sequence of measurable sets. Furthermore,

$$\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} (A_1 - A_n) = A_1 - \bigcap_{n=1}^{\infty} A_n. \quad (1.18)$$

Claim 1.1: If A, B are measurable sets with $A \subseteq B$ and $\mu(A) < \infty$, then $\mu(B - A) = \mu(B) - \mu(A)$.

Proof. $B = A \cup (B - A)$ and this is a disjoint decomposition. Therefore,

$$\mu(B) = \mu(A) + \mu(B - A). \quad (1.19)$$

Rearranging gives the desired result. $\square$

By applying part (a), we get

$$\mu \left(\bigcup_{n=1}^{\infty} B_n \right) = \lim_{n \rightarrow \infty} \mu(B_n). \quad (1.20)$$

Applying [Claim 1.1](#) into both sides of the equation, we get

$$\mu(A_1) - \mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \mu(A_1) - \lim_{n \rightarrow \infty} \mu(A_n). \quad (1.21)$$

■

Remark 1.7. The condition that $\mu(A_1) < \infty$ (or at least $\mu(A_n) < \infty$ for some n) is important here. For example, consider

$$\Omega = \mathbb{N}, \quad \Sigma = \mathcal{P}(\mathbb{N}), \quad \mu = \text{counting measure}.$$

Here, $A_n = \{n, n+1, n+2, \dots\}$ is a decreasing sequence with $\mu(A_n) = \infty$ for every n . But

$$\mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \mu(\emptyset) = 0 \text{ and } \lim_{n \rightarrow \infty} \mu(A_n) = \infty. \quad (1.22)$$

Example 1.6

Lebesgue measure on $\mathbb{R}^d$ is not finite (the whole space has infinite volume), but it *is* σ -finite: taking $A_n = B(0, n)$, the ball of radius n centered at the origin, each A_n has finite volume and $\mathbb{R}^d = \bigcup_{n=1}^{\infty} A_n$.

Remark 1.8. The choice of Ω , Σ , and μ is entirely up to the application at hand. There is no canonical choice in general. However, certain combinations are used so pervasively that they become the default setting for the theory; the main example developed in this course will be $\mathbb{R}^d$ equipped with the Borel (or Lebesgue) σ -algebra and Lebesgue measure.

Definition 1.5 (Measurable function). Let (Ω, Σ) be a measurable space. A function $f : \Omega \rightarrow \mathbb{R}$ is called **measurable** (with respect to Σ) if, for every interval $I \subseteq \mathbb{R}$,

$$f^{-1}(I) := \{x \in \Omega : f(x) \in I\} \in \Sigma. \quad (1.23)$$

Remark 1.9. It suffices to check this condition for intervals of the special form $I = (a, \infty)$, $a \in \mathbb{R}$: since Σ is closed under complements, countable unions, and countable intersections, every interval (open, closed, half-open, bounded, or unbounded) can be built from intervals of this one type. Because,

$$\begin{aligned} (-\infty, a] &= (a, \infty)^c, \quad [a, \infty) = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, \infty\right), \quad (-\infty, a) = [a, \infty)^c, \\ (a, b) &= (a, \infty) \cap (-\infty, b), \quad [a, b] = [a, \infty) \cap (-\infty, b], \\ [a, b) &= [a, \infty) \cap (-\infty, b), \quad (a, b] = (a, \infty) \cap (-\infty, b]; \end{aligned}$$

and preimage distributes union, intersections and complements. That's why many texts simply *define* measurability by requiring $f^{-1}((a, \infty)) \in \Sigma$ for all $a \in \mathbb{R}$.

Remark 1.10 (Measurability depends on the choice of Σ). Whether a given function is measurable depends entirely on the σ -algebra chosen on the domain. It is not an intrinsic property of the function alone.

Example 1.7

Let $\Omega = \mathbb{R}$ and $\Sigma = \{\emptyset, \mathbb{R}\}$ (the trivial σ -algebra).

- The constant function $f(x) = 1$ is measurable: for any interval I , $f^{-1}(I)$ is either $\mathbb{R}$ (if $1 \in I$) or $\emptyset$ (if $1 \notin I$), and both lie in Σ .
- The identity function $f(x) = x$ is **not** measurable: for a bounded interval such as $I = (0, 1)$, $f^{-1}(I) = (0, 1) \notin \{\emptyset, \mathbb{R}\}$.

This illustrates that measurability is relative to Σ : the same function can be measurable for one σ -algebra and not for another.

Example 1.8 (Basic examples of measurable functions)

The following are basic examples of measurable functions (with respect to the Borel σ -algebra, unless otherwise specified):

- Any constant function $f \equiv c$ is measurable.
- For a measurable set $A \subseteq \Omega$, the **characteristic function** of A is defined by

$$\chi_A(x) := \begin{cases} 1, & \text{if } x \in A; \\ 0, & \text{if } x \notin A. \end{cases} \quad (1.24)$$

This function is measurable: for $a \in \mathbb{R}$, the following sets are all measurable

$$\{x \in \Omega : \chi_A(x) > a\} = \begin{cases} \Omega, & a < 0; \\ A, & 0 \leq a < 1; \\ \emptyset, & a \geq 1. \end{cases} \quad (1.25)$$

- Every continuous function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is measurable with respect to the Borel σ -algebra: for any $a \in \mathbb{R}$, the set $\{x : f(x) > a\} = f^{-1}((a, \infty))$ is open (being the preimage of an open set under a continuous map), hence Borel measurable.
- Every monotone function $f : \mathbb{R} \rightarrow \mathbb{R}$ (increasing or decreasing) is measurable with respect to the Borel σ -algebra: for any $a \in \mathbb{R}$, the set $\{x : f(x) > a\}$ is always an interval (a ray, of one of the forms (c, ∞) , $[c, \infty)$, $(-\infty, c)$, or $(-\infty, c]$, depending on monotonicity and endpoint conventions), and every interval is Borel measurable.

Remark 1.11 (Measurable maps between measure spaces). More generally, if (Ω_1, Σ_1) and (Ω_2, Σ_2) are two measurable spaces, a function $f : \Omega_1 \rightarrow \Omega_2$ is called **measurable** if $f^{-1}(B) \in \Sigma_1$ for every $B \in \Sigma_2$. The definition above is the special case $\Omega_2 = \mathbb{R}$ with $\Sigma_2 = \mathcal{B}(\mathbb{R})$, the Borel σ -algebra.

It will be convenient for us to also allow our functions to take values in the extended real line $\overline{\mathbb{R}}$. The definition is the same as before: a function $f : \Omega \rightarrow \overline{\mathbb{R}}$ is **measurable** if the preimage of any extended interval is measurable, or, alternatively, if the sets $\{x \in \Omega : f(x) > a\}$ are measurable for each $a \in \mathbb{R}$. In this case, observe that the sets

$$\{x \in \Omega : f(x) = +\infty\} = \bigcap_{n=1}^{\infty} \{x \in \Omega : f(x) > n\}, \quad (1.26)$$

$$\{x \in \Omega : f(x) = -\infty\} = \left(\bigcup_{n=1}^{\infty} \{x \in \Omega : f(x) > -n\} \right)^c \quad (1.27)$$

are both measurable.

We shall adopt the convention that

$$0 \cdot (\pm\infty) = 0 \text{ and } (\pm\infty) + \pm(\mp\infty) = 0. \quad (1.28)$$

Let $\mathcal{M} = \mathcal{M}(\Omega, \Sigma)$ denote the set of measurable functions $f : \Omega \rightarrow \mathbb{R}$ (or $\overline{\mathbb{R}}$).

Proposition 1.5

Let $f, g \in \mathcal{M}$ and $c \in \mathbb{R}$. Then each of the following functions is also measurable:

- (a) cf ;
- (b) $f \pm g$;
- (c) fg ;
- (d) $\frac{f}{g}$, if $g \neq 0$ everywhere;
- (e) $|f|$.

Proof. We are going to use the trick that f is measurable if the sets $\{x \in \Omega : f(x) > a\}$ are all measurable.

- (a) If $c = 0$, then $cf = 0$, which is a constant function, so it's measurable. For $c > 0$,

$$\{x \in \Omega : c \cdot f(x) > a\} = \left\{x \in \Omega : f(x) > \frac{a}{c}\right\} \quad (1.29)$$

is measurable since f is a measurable function. For $c < 0$,

$$\{x \in \Omega : c \cdot f(x) > a\} = \left\{x \in \Omega : f(x) < \frac{a}{c}\right\} \quad (1.30)$$

is measurable since f is a measurable function.

(b) Note that (assuming $f(x), g(x)$ are finite)

$$\begin{aligned} f(x) + g(x) > a &\iff f(x) > a - g(x) \\ &\iff \exists q \in \mathbb{Q} \text{ such that } f(x) > q > a - g(x) \\ &\iff \exists q \in \mathbb{Q} \text{ such that } f(x) > q \text{ and } g(x) > a - q. \end{aligned}$$

$$\begin{aligned} &\{x \in \Omega : f(x) + g(x) > a\} \\ &= \{x \in \Omega : \text{both } f(x) \text{ and } g(x) \text{ are finite, and } f(x) + g(x) > a\} \\ &\quad \cup (\{x \in \Omega : f(x) = \infty\} \cap \{x \in \Omega : g(x) > -\infty\}) \\ &\quad \cup (\{x \in \Omega : g(x) = \infty\} \cap \{x \in \Omega : f(x) > -\infty\}) \\ &\quad \cup X, \end{aligned} \tag{1.31}$$

where X is

$$X = \begin{cases} \{x \in \Omega : f(x) = +\infty, g(x) = -\infty\} \cup \{x \in \Omega : g(x) = +\infty, f(x) = -\infty\}, & a < 0, \\ \emptyset, & a \geq 0. \end{cases} \tag{1.32}$$

Therefore,

$$\begin{aligned} &\{x \in \Omega : f(x) + g(x) > a\} \\ &= \bigcup_{q \in \mathbb{Q}} (\{x \in \Omega : f(x) > q\} \cap \{x \in \Omega : g(x) > a - q\} \cap \{x \in \Omega : f(x), g(x) \in \mathbb{R}\}) \\ &\quad \cup (\{x \in \Omega : f(x) = \infty\} \cap \{x \in \Omega : g(x) > -\infty\}) \\ &\quad \cup (\{x \in \Omega : g(x) = \infty\} \cap \{x \in \Omega : f(x) > -\infty\}) \\ &\quad \cup X. \end{aligned} \tag{1.33}$$

This is a measurable set since it's made up of measurable sets by taking union and intersection.

(c) We're gonna prove an intermediate lemma first.

Claim 1.2: If f is measurable, then so is f^2 .

Proof. If $a < 0$, then $\{x \in \Omega : f(x)^2 > a\} = \Omega$, is measurable. For $a \geq 0$,

$$\{x \in \Omega : f(x)^2 > a\} = \{x \in \Omega : f(x) > \sqrt{a}\} \cup \{x \in \Omega : f(x) < -\sqrt{a}\} \tag{1.34}$$

is measurable. □

Now notice that

$$fg = \frac{1}{4} [(f+g)^2 - (f-g)^2], \tag{1.35}$$

is indeed measurable.

(d) It suffices to show that $\frac{1}{g}$ is measurable.

$$\left\{x \in \Omega : \frac{1}{g(x)} > a\right\} = \begin{cases} \{x \in \Omega : 0 < g(x) < \frac{1}{a}\}, & a > 0, \\ \{x \in \Omega : g(x) > 0\}, & a = 0, \\ \{x \in \Omega : \frac{1}{a} < g(x) < 0\}, & a < 0. \end{cases} \quad (1.36)$$

These are all measurable since g is measurable.

(e) If $a < 0$, then $\{x \in \Omega : |f(x)| > a\} = \Omega$, is measurable. For $a \geq 0$,

$$\{x \in \Omega : |f(x)| > a\} = \{x \in \Omega : f(x) > a\} \cup \{x \in \Omega : f(x) < -a\} \quad (1.37)$$

is measurable. ■

Proposition 1.6

Let $(f_n)_{n=1}^\infty$ be a countable sequence of measurable functions $\Omega \rightarrow \overline{\mathbb{R}}$. Then each of the following functions is measurable:

- (a) $f(x) = \sup_n f_n(x)$;
- (b) $f(x) = \inf_n f_n(x)$;
- (c) $f(x) = \limsup_{n \rightarrow \infty} f_n(x)$;
- (d) $f(x) = \liminf_{n \rightarrow \infty} f_n(x)$.

Consequently, if $f_n \rightarrow f$ pointwise, then f is measurable.

Proof. (a) Note that

$$\sup_n f_n(x) > a \iff f_n(x) > a \text{ for some } n. \quad (1.38)$$

Therefore,

$$\left\{x \in \Omega : \sup_n f_n(x) > a\right\} = \bigcup_n \{x \in \Omega : f_n(x) > a\}, \quad (1.39)$$

is measurable.

(b) Note that

$$\inf_n f_n(x) = - \left(\sup_n (-f_n(x)) \right). \quad (1.40)$$

(c) Follows from the fact that

$$\limsup_{n \rightarrow \infty} f_n(x) = \inf_m \sup_{n \geq m} f_n(x). \quad (1.41)$$

(d) Follows from the fact that

$$\liminf_{n \rightarrow \infty} f_n(x) = \sup_m \inf_{n \geq m} f_n(x). \quad (1.42)$$

When $f_n \rightarrow f$ pointwise, then the limit, limsup and liminf coincides. ■

Remark 1.12. In the case of complex-valued functions, we say that $f : \Omega \rightarrow \mathbb{C}$ is measurable if its real part and imaginary parts are both measurable.

Definition 1.6 (Almost everywhere). Let (Ω, Σ, μ) be a measure space. We say that a property (P) holds μ -**almost everywhere** (or simply **almost everywhere**, when the measure μ is clear from context), abbreviated μ -a.e. or a.e., if there exists a set $N \in \Sigma$ with $\mu(N) = 0$ such that (P) holds on N^c . Such a set N is called a **null set**.

Example 1.9

Typical almost-everywhere statements include:

- $f = g$ almost everywhere, meaning $\{x \in \Omega : f(x) \neq g(x)\}$ is contained in a null set;
- $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ almost everywhere, meaning the set of x for which this limit fails to hold is contained in a null set.

Remark 1.13 (Subsets of null sets need not be measurable). If $\mu(N) = 0$ and $B \subseteq N$, one would like to conclude that $\mu(B) = 0$ as well. However, this conclusion presupposes $B \in \Sigma$, which need not be the case, since Σ may fail to contain every subset of a null set.

Definition 1.7 (Complete measure). A measure μ on (Ω, Σ) is called **complete** if its domain Σ contains every subset of every μ -null set; that is, whenever $N \in \Sigma$ satisfies $\mu(N) = 0$ and $B \subseteq N$, then $B \in \Sigma$ (and consequently $\mu(B) = 0$).

Remark 1.14. Every measure can be made complete by slightly enlarging its σ -algebra, as follows. Let (Ω, Σ, μ) be a measure space. Let

$$\mathcal{N} := \{N \in \Sigma : \mu(N) = 0\}, \quad \bar{\Sigma} := \{A \cup B : A \in \Sigma \text{ and } B \subseteq N \text{ for some } N \in \mathcal{N}\}. \quad (1.43)$$

$\bar{\Sigma}$ is a σ -algebra. Indeed:

1. Take $A \cup B \in \bar{\Sigma}$, where $A \in \Sigma$ and $B \subseteq N$ for some $N \in \mathcal{N}$. Then

$$(A \cup B)^c = A^c \cap B^c = A^c - (B \cap A^c). \quad (1.44)$$

We're gonna use $W - U = (W - V) \cup (V - U)$ for $U \subseteq V \subseteq W$, with $U = B \cap A^c$, $V = N \cap A^c$, $W = A^c$. Then the RHS of (1.44) becomes,

$$[A^c - (N \cap A^c)] \cup [(N \cap A^c) - (B \cap A^c)]. \quad (1.45)$$

$A^c - (N \cap A^c) \in \Sigma$ since it's made up of elements of Σ using complementation and finite intersection. $(N \cap A^c) - (B \cap A^c) \subseteq N$. Therefore, $(A \cup B)^c \in \bar{\Sigma}$.

2. Let $A_i \cup B_i \in \bar{\Sigma}$ be a countable sequence of sets, with $A_i \in \Sigma$ and $B_i \subseteq N_i$ with $\mu(N_i) = 0$. Then

$$\mu\left(\bigcup_{i=1}^{\infty} N_i\right) \leq \sum_{i=1}^{\infty} \mu(N_i) = 0, \quad (1.46)$$

implying that $\bigcup_{i=1}^{\infty} N_i \in \mathcal{N}$. Now,

$$\bigcup_{i=1}^{\infty} (A_i \cup B_i) = \left(\bigcup_{i=1}^{\infty} A_i\right) \cup \left(\bigcup_{i=1}^{\infty} B_i\right), \quad (1.47)$$

where $\bigcup_{i=1}^{\infty} A_i \in \Sigma$, and $\bigcup_{i=1}^{\infty} B_i \subseteq \bigcup_{i=1}^{\infty} N_i \in \mathcal{N}$.

There is a unique extension $\bar{\mu}$ of μ to a complete measure on $\bar{\Sigma}$, given by

$$\bar{\mu}(A \cup B) = \mu(A), \quad (1.48)$$

where $A \in \Sigma$ and $B \subseteq N$ for some $N \in \mathcal{N}$. The measure space $(\Omega, \bar{\Sigma}, \bar{\mu})$ is called the **completion** of (Ω, Σ, μ) .

Proposition 1.7

$\bar{\mu}$ is a complete measure, and it's the unique measure on $\bar{\Sigma}$ that extends μ .

Proof. Suppose $\mu(A \cup B) = 0$ for some $A \cup B \in \bar{\Sigma}$, where $A \in \Sigma$ and $B \subseteq N$ with $\mu(N) = 0$. Choose any subset $C \subseteq A \cup B$. We can write it as

$$C = \emptyset \cup C, \text{ for } C \subseteq A \cup N, \quad (1.49)$$

where $A \cup N \in \Sigma$ with $\mu(A \cup N) = 0$. Therefore, $C \in \bar{\Sigma}$, and hence $\bar{\mu}$ is a complete measure.

Assume for the sake of contradiction that μ' is a measure on $\bar{\Sigma}$ that extends μ , i.e. $\mu'|_{\Sigma} = \mu$. For $A \cup B \in \bar{\Sigma}$, where $A \in \Sigma$ and $B \subseteq N$ with $\mu(N) = 0$, we have

$$\bar{\mu}(A \cup B) = \mu(A) = \mu'(A) \leq \mu'(A \cup B) \leq \mu'(A \cup N) = \mu(A \cup N) = \mu(A) = \bar{\mu}(A \cup B). \quad (1.50)$$

Therefore, $\mu' = \bar{\mu}$. ■

Lemma 1.8

The following implications hold **if and only if** the measure μ is complete:

- (i) if f is measurable and $f = g$ μ -a.e., then g is measurable;
- (ii) if $(f_n)_{n \geq 1}$ are measurable and $f_n \rightarrow f$ μ -a.e., then f is measurable.

Proof. (i) Suppose μ is complete; and f is measurable and $f = g$ a.e. Then $f \neq g$ on $B \subseteq N$, where $\mu(N) = 0$. Since μ is complete, $B \in \Sigma$. Now,

$$\{x \in \Omega : g(x) > a\} = (\{x \in \Omega : g(x) > a\} \cap B) \cup (\{x \in \Omega : g(x) > a\} \cap B^c) \quad (1.51)$$

$$= (\text{subset of a null-set}) \cup (\{x \in \Omega : f(x) > a\} \cap B^c), \quad (1.52)$$

which is a measurable set.

Conversely, suppose that whenever f is measurable and $f = g$ a.e., we have that g is measurable. Now, let $N \in \mathcal{N}$ and $B \subseteq N$. Take

$$f \equiv 0, \quad g = \chi_B \quad (1.53)$$

They agree on N^c , so $f = g$ a.e. f is a constant function so it's measurable. Therefore, g is also measurable, which forces

$$g^{-1}\{1\} = B \in \Sigma. \quad (1.54)$$

(ii) Suppose $f_n \rightarrow g$ (everywhere). Then g is measurable, and $f = g$ a.e. Therefore, f is measurable if and only if μ is complete. ■

Lemma 1.9 (Borel–Cantelli lemma)

Let (Ω, Σ, μ) be a measure space. Let $(E_k)_{k=1}^\infty$ be a countable family of measurable sets of Ω such that

$$\sum_{k=1}^{\infty} \mu(E_k) < \infty. \quad (1.55)$$

Let $E := \{x \in \Omega : x \in E_k \text{ for infinitely many } k\}$. Then E is measurable and $\mu(E) = 0$.

Proof. Note that

$$\begin{aligned} x \in E_k \text{ for infinitely many } k &\iff \forall n \exists m \geq n \text{ such that } x \in E_m \\ &\iff \forall n, x \in \bigcup_{m \geq n} E_m \\ &\iff x \in \bigcap_{n=1}^{\infty} \bigcup_{m \geq n} E_m. \end{aligned}$$

Therefore, $E = \bigcap_{n=1}^{\infty} \bigcup_{m \geq n} E_m$ is measurable as it's made up of measurable sets by countable unions and intersections. Now,

$$\mu\left(\bigcup_{m \geq 1} E_m\right) \leq \sum_{m=1}^{\infty} \mu(E_m) < \infty. \quad (1.56)$$

By continuity, since $\left\{\bigcup_{m \geq n} E_m\right\}_n$ is a decreasing sequence,

$$\mu(E) = \lim_{n \rightarrow \infty} \mu\left(\bigcup_{m \geq n} E_m\right) \leq \lim_{n \rightarrow \infty} \sum_{m=n}^{\infty} \mu(E_m) = 0. \quad (1.57)$$

■

Exercise 1.2. Let $(f_n)_{n \geq 1}$ be a sequence of (measurable) functions on a measure space (Ω, Σ, μ) . Suppose that for every $x \in \Omega$, the sequence $(f_n(x))_{n \geq 1}$ is bounded. Let

$$E := \{x \in \Omega : (f_n(x))_{n \geq 1} \text{ converges}\}. \quad (1.58)$$

Show that $E \in \Sigma$.

Solution. $(f_n(x))_{n \geq 1}$ converges if and only if $\limsup_n f_n(x) = \liminf_n f_n(x)$. We have already shown that $\limsup_n f_n$ and $\liminf_n f_n$ are measurable. Therefore, so is $h := \limsup_n f_n - \liminf_n f_n$. Now,

$$E = \{x \in \Omega : (f_n(x))_{n \geq 1} \text{ converges}\} = h^{-1}\{0\}, \quad (1.59)$$

which is measurable since h is measurable. ■

Exercise 1.3. For each $n \geq 1$, let $f_n : [0, 1] \rightarrow \mathbb{R}$ be a continuous function. Suppose that:

- (i) $0 \leq f_n(x)$ for all n and all $x \in [0, 1]$;
- (ii) the sequence of partial sums $\sum_{n=1}^N f_n(x)$ converges uniformly on $[0, 1]$ as $N \rightarrow \infty$;
- (iii) the sum $\sum_{n=1}^{\infty} (\sup\{f_n(x) : 0 \leq x \leq 1\})$ diverges.

Either give an example of such a sequence of functions, or prove that no such sequence exists.

Solution. One such sequence of functions is

$$f_n(x) = \begin{cases} \frac{1}{n} & \text{if } x = \frac{1}{n}, \\ 0 & \text{otherwise.} \end{cases} \quad (1.60)$$

Then

$$\sum_{n=1}^{\infty} (\sup\{f_n(x) : 0 \leq x \leq 1\}) = \sum_{n=1}^{\infty} \frac{1}{n} \text{ diverges.} \quad (1.61)$$

$$\sum_{n=1}^N f_n(x) = \begin{cases} x & \text{if } x = \frac{1}{n} \text{ for } 1 \leq n \leq N, \\ 0 & \text{otherwise,} \end{cases} \quad (1.62)$$

which converges uniformly to

$$f(x) = \begin{cases} x & \text{if } x = \frac{1}{n} \text{ for } n = 1, 2, 3, \dots, \\ 0 & \text{otherwise,} \end{cases} \quad (1.63)$$

■

§2 11 September 2026

Having built the machinery of σ -algebras, measures, and measurable functions, we now turn to the central question of this part of the course:

Given a measure space (Ω, Σ, μ) and a measurable function f , how do we define $\int_{\Omega} f \, d\mu$?

The construction proceeds in three stages: first for characteristic functions, then for simple functions, and finally for general non-negative measurable functions (by approximation). Integration of signed or complex-valued functions is deferred to a later lecture.

Definition 2.1 (Characteristic function). Let $A \in \Sigma$ be a measurable set. The **characteristic function** (or **indicator function**) of A is the function $\chi_A : \Omega \rightarrow \mathbb{R}$ (also written $\mathbf{1}_A$) defined by

$$\chi_A(x) := \begin{cases} 1, & x \in A; \\ 0, & x \notin A. \end{cases} \quad (2.1)$$

Remark 2.1. As noted already, χ_A is always measurable when $A \in \Sigma$: for an interval $I \subseteq \mathbb{R}$,

$$\chi_A^{-1}(I) = \begin{cases} \Omega, & 0, 1 \in I; \\ A, & 1 \in I, 0 \notin I; \\ A^c, & 0 \in I, 1 \notin I; \\ \emptyset, & 0, 1 \notin I, \end{cases} \quad (2.2)$$

and each of $\Omega, A, A^c, \emptyset$ lies in Σ .

The entire theory of integration is built by insisting on one basic axiom, motivated by the idea that the integral should compute the “volume” below the graph of a function: for a measurable set A and a constant height $c \geq 0$, the volume of the block of height c over the base A should be c times the size of the base. This gives the defining relation

$$\int_{\Omega} c \chi_A \, d\mu := c \mu(A), \quad c \geq 0, \, A \in \Sigma, \quad (2.3)$$

and, in particular (taking $c = 1$),

$$\int_{\Omega} \chi_A \, d\mu := \mu(A). \quad (2.4)$$

Definition 2.2 (Simple function). A measurable function $\varphi : \Omega \rightarrow \mathbb{R}$ is called **simple** if it takes only finitely many values.

Remark 2.2 (Standard representation). If φ is simple and $a_1, \dots, a_N$ are its distinct *non-zero* values, set

$$A_i := \varphi^{-1}(\{a_i\}), \quad i = 1, \dots, N. \quad (2.5)$$

Each A_i is measurable (since φ is measurable), the sets $A_1, \dots, A_N$ are pairwise disjoint, and

$$\varphi = \sum_{i=1}^N a_i \chi_{A_i}. \quad (2.6)$$

This expression is called the **standard representation** of φ .

Remark 2.3 (Non-standard representations). A simple function can typically be written as a finite linear combination of characteristic functions in more than one way, using sets that are *not* disjoint and coefficients that are *not* distinct. For example, on disjoint sets A, B, C with

$$\varphi = 5\chi_A + 8\chi_B + 3\chi_C \quad (2.7)$$

(the standard representation), one may check directly that

$$\varphi = 5\chi_{A \cup B} + 3\chi_{B \cup C} \quad (2.8)$$

as well, since on A this gives 5, on B it gives $5 + 3 = 8$, and on C it gives 3 — matching the standard representation, even though $A \cup B$ and $B \cup C$ overlap.

Definition 2.3 (Integral of a non-negative simple function). Let φ be a non-negative simple function with standard representation $\varphi = \sum_{i=1}^N a_i \chi_{A_i}$ (so $a_i > 0$ and the A_i are pairwise disjoint, possibly with $\mu(A_i) = \infty$). We define

$$\int_{\Omega} \varphi \, d\mu := \sum_{i=1}^N a_i \mu(A_i) \in [0, \infty]. \quad (2.9)$$

Lemma 2.1 (Independence of the representation)

Let $\varphi \geq 0$ be a simple function, and suppose

$$\varphi = \sum_{n=1}^N a_n \chi_{A_n} = \sum_{m=1}^M b_m \chi_{B_m} \quad (2.10)$$

are two representations of φ as finite linear combinations of characteristic functions, where the sets A_n (resp. B_m) need *not* be pairwise disjoint, and the coefficients $a_n, b_m \in \mathbb{R}$ need *not* be distinct or non-negative (only φ itself is assumed non-negative; we adopt the convention $0 \times \infty = 0$ if some $\mu(A_n) = \infty$ with $a_n = 0$). Then

$$\sum_{n=1}^N a_n \mu(A_n) = \sum_{m=1}^M b_m \mu(B_m). \quad (2.11)$$

Proof. We'll show that $\sum_{n=1}^N a_n \mu(A_n) = \sum_{i=1}^K r_i \mu(E_i)$, where $\varphi = \sum_{i=1}^K r_i \chi_{E_i}$ is the standard representation.

Case 1: A_i 's are pairwise disjoint, but a_i 's need not be distinct. In this case, for each $a \in \{a_1, \dots, a_N\}$, set $B_a = \bigcup_{a_i=a} A_i$. Then it's clear that $B_a \cap B_{a'} = \emptyset$, since A_i 's are pairwise disjoint. Furthermore, $\varphi^{-1}(a) = B_a$. Therefore, $\varphi = \sum_a a \chi_{B_a}$ is the standard representation.

$$\sum_a a \mu(B_a) = \sum_a a \sum_{a_i=a} \mu(A_i) = \sum_{i=1}^N a_i \mu(A_i). \quad (2.12)$$

Case 2: Now let's assume that A_i 's are not necessarily pairwise disjoint.

Claim 2.1: There exists a refinement of the set $\bigcup_{i=1}^N A_i$ with sets $B_1, \dots, B_M$ such that $\bigcup_{i=1}^N A_i = \bigcup_{j=1}^M B_j$, and B_j 's are pairwise disjoint, and for each i , $A_i = \bigcup_{B_j \subseteq A_i} B_j$.

Proof. For each nonempty subset S of $\{1, \dots, N\}$, we set

$$B_S = \left(\bigcap_{i \in S} A_i \right) \cap \left(\bigcap_{i \notin S} A_i^c \right). \quad (2.13)$$

This gives a collection of $2^N - 1$ pairwise disjoint sets. Now, we'll show that

$$A_i = \bigcup_{B_S \subseteq A_i} B_S. \quad (2.14)$$

Clearly, $\bigcup_{B_S \subseteq A_i} B_S \subseteq A_i$ holds. Now, take $x \in A_i$. Let $S' = \{i : x \in A_i\}$, which is a nonempty set. Then $x \in B_{S'} \subseteq A_i$ since S' contains i . Therefore, $\bigcup_{B_S \subseteq A_i} B_S \supseteq A_i$.

This proves $\bigcup_{S \neq \emptyset} B_S \supseteq \bigcup_{i=1}^N A_i$. The reverse containment holds since $B_S \subseteq \bigcup_{i=1}^N A_i$. $\square$

Now, for each S , we define $b_S = \sum_{A_i \supseteq B_S} a_i$. Then

$$\sum_{i=1}^N a_i \mu(A_i) = \sum_{i=1}^N a_i \sum_{B_S \subseteq A_i} \mu(B_S) = \sum_S \sum_{B_S \subseteq A_i} a_i \mu(B_S) = \sum_S b_S \mu(B_S). \quad (2.15)$$

This boils down to case 1, and we know that this is equal to the sum in the standard representation. $\blacksquare$

Remark 2.4. The integral of a non-negative simple function may be computed using *any* representation of φ as a finite linear combination of characteristic functions of measurable sets, not only the standard one. Equivalently, a non-negative simple function may be described as any finite non-negative linear combination of characteristic functions of measurable sets, and its integral is unambiguously

$$\int_{\Omega} \varphi d\mu = \sum_n a_n \mu(A_n) \quad (2.16)$$

for any such representation $\varphi = \sum_n a_n \chi_{A_n}$.

Lemma 2.2

Let φ and ψ be simple functions in $\mathcal{M}^+$.

- (a) If $c \in \mathbb{R}$, then $\int c\varphi \, d\mu = c \int \varphi \, d\mu$.
- (b) $\int (\varphi + \psi) \, d\mu = \int \varphi \, d\mu + \int \psi \, d\mu$.
- (c) If $\varphi \leq \psi$, then $\int \varphi \, d\mu \leq \int \psi \, d\mu$.
- (d) $A \mapsto \int_A \varphi \, d\mu := \int_{\Omega} \varphi \cdot \chi_A \, d\mu$ is a measure on Σ .

Proof. (a) Let $\varphi = \sum_{i=1}^n a_i \chi_{A_i}$. Then $c\varphi = \sum_{i=1}^n ca_i \chi_{A_i}$.

$$\int_{\Omega} c\varphi \, d\mu = \sum_{i=1}^n ca_i \mu(A_i) = c \sum_{i=1}^n a_i \mu(A_i) = c \int_{\Omega} \varphi \, d\mu. \quad (2.17)$$

(b) Let $\varphi = \sum_{i=1}^n a_i \chi_{A_i}$ and $\psi = \sum_{j=1}^m b_j \chi_{B_j}$. Then $\varphi + \psi = \sum_{i=1}^n a_i \chi_{A_i} + \sum_{j=1}^m b_j \chi_{B_j}$.

$$\int_{\Omega} (\varphi + \psi) \, d\mu = \sum_{i=1}^n a_i \mu(A_i) + \sum_{j=1}^m b_j \mu(B_j) = \int_{\Omega} \varphi \, d\mu + \int_{\Omega} \psi \, d\mu. \quad (2.18)$$

(c) $\psi - \varphi$ is a non-negative simple function, whose integral is non-negative, by definition.

(d) Let $\varphi = \sum_{i=1}^n a_i \chi_{E_i}$ be the standard representation.

$$\nu(A) := \int_{\Omega} \varphi \cdot \chi_A \, d\mu = \int_{\Omega} \sum_{i=1}^n a_i \chi_{E_i} \cdot \chi_A \, d\mu = \int_{\Omega} \sum_{i=1}^n a_i \chi_{E_i \cap A} \, d\mu = \sum_{i=1}^n a_i \mu(E_i \cap A). \quad (2.19)$$

Clearly, $\nu(A) \geq 0$ and $\nu(\emptyset) = 0$. Now, if $\{A_n\}_n$ is a countable collection of pairwise disjoint sets,

$$\nu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{i=1}^n a_i \mu\left(E_i \cap \bigcup_{n=1}^{\infty} A_n\right) = \sum_{i=1}^n a_i \mu\left(\bigcup_{n=1}^{\infty} (E_i \cap A_n)\right) \quad (2.20)$$

$E_i \cap A_n$'s are all pairwise disjoint, so

$$\nu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{i=1}^n a_i \sum_{n=1}^{\infty} \mu(E_i \cap A_n) = \sum_{n=1}^{\infty} \sum_{i=1}^n a_i \mu(E_i \cap A_n) = \sum_{n=1}^{\infty} \nu(E_i \cap A_n). \quad (2.21)$$

■

We now turn to general non-negative measurable functions. The key technical tool, used repeatedly throughout analysis, is to approximate an object in a large class (here, non-negative measurable functions) by objects in a smaller class we already understand (here, simple functions).

Theorem 2.3 (Approximation by simple functions)

Let $f : \Omega \rightarrow [0, \infty]$ be measurable. Then there exists a sequence of simple functions $(\varphi_n)_{n \geq 1}$, each satisfying $0 \leq \varphi_n \leq \varphi_{n+1}$, such that

$$\varphi_n(x) \longrightarrow f(x) \quad \text{for every } x \in \Omega. \quad (2.22)$$

Proof. For each $n \geq 1$ and each integer k with $0 \leq k \leq n2^n - 1$, define the measurable set

$$A_{k,n} := \left\{ x \in \Omega : \frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n} \right\}, \quad (2.23)$$

and, to handle the top of the range, set

$$A_{n2^n,n} := \{x \in \Omega : f(x) \geq n\}. \quad (2.24)$$

These sets are measurable (being preimages of intervals under f) and, for each fixed n , pairwise disjoint, with

$$\Omega = \bigcup_{k=0}^{n2^n} A_{k,n}. \quad (2.25)$$

Define

$$\varphi_n := \sum_{k=0}^{n2^n} \frac{k}{2^n} \chi_{A_{k,n}}, \quad (2.26)$$

a non-negative simple function. In words: φ_n rounds f down to the nearest multiple of 2^{-n} , on the region where $f < n$, and sets $\varphi_n = n$ where $f \geq n$.

Monotonicity. Passing from level n to level $n+1$ refines each interval $[k/2^n, (k+1)/2^n)$ into

$$\left[\frac{k}{2^n}, \frac{k+1}{2^n} \right) = \left[\frac{2k}{2^{n+1}}, \frac{2k+1}{2^{n+1}} \right) \cup \left[\frac{2k+1}{2^{n+1}}, \frac{2k+2}{2^{n+1}} \right). \quad (2.27)$$

So $A_{k,n} = A_{2k,n+1} \cup A_{2k+1,n+1}$; on either piece, the assigned value at level $n+1$ is at least the value $k/2^n$ assigned at level n . Likewise, the top set $A_{n2^n,n} = \{f \geq n\}$ splits, at level $n+1$, into the part where $n \leq f < n+1$ (now assigned a value $\geq n$, refined further) and the part where $f \geq n+1$ (assigned the value $n+1 > n$). Hence $\varphi_n \leq \varphi_{n+1}$ pointwise.

Pointwise convergence. Fix $x \in \Omega$. If $f(x) = \infty$, then $\varphi_n(x) = n \rightarrow \infty = f(x)$. If $f(x) < \infty$, then once $n > f(x)$, we have $x \in A_{k,n}$ for the unique k with

$$\frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n}, \quad (2.28)$$

so

$$0 \leq f(x) - \varphi_n(x) < \frac{1}{2^n} \longrightarrow 0. \quad (2.29)$$

Hence $\varphi_n(x) \rightarrow f(x)$ for every $x \in \Omega$. ■

Remark 2.5. The convergence in this proposition holds **everywhere**, not merely almost everywhere, and the approximating sequence is **monotonically increasing**. Both features will be essential in the proof of the Monotone Convergence Theorem below.

Definition 2.4 (Integral of a non-negative measurable function). Let $f : \Omega \rightarrow [0, \infty]$ be measurable. We define

$$\int_{\Omega} f \, d\mu := \sup \left\{ \int_{\Omega} \varphi \, d\mu : \varphi \text{ simple, } 0 \leq \varphi \leq f \right\} \in [0, \infty]. \quad (2.30)$$

Remark 2.6. This definition could have been adopted directly, without first proving the approximation proposition above; the proposition simply confirms that the class of simple functions lying below f is rich enough to be useful (e.g. in proving the Monotone Convergence Theorem).

Notation for the integral varies across the literature and is otherwise a matter of convention: one may write $\int_{\Omega} f \, d\mu$, $\int_{\Omega} f(x) \, d\mu(x)$, $\int f \, d\mu$ (when Ω is clear), or $\int f$ (when μ is also clear); some authors (especially in physics) write the measure before the function. Any of these is acceptable as long as the underlying set, σ -algebra, and measure are unambiguous.

The following two properties follow immediately from the definition as a supremum.

Proposition 2.4 (Monotonicity)

If $f, g : \Omega \rightarrow [0, \infty]$ are measurable and $f \leq g$, then

$$\int_{\Omega} f \, d\mu \leq \int_{\Omega} g \, d\mu. \quad (2.31)$$

Proof. Every simple function φ with $0 \leq \varphi \leq f$ also satisfies $0 \leq \varphi \leq g$, so the set over which the supremum defining $\int_{\Omega} f \, d\mu$ is taken is contained in the set over which the supremum defining $\int_{\Omega} g \, d\mu$ is taken. A supremum over a smaller set cannot exceed a supremum over a larger set. ■

Proposition 2.5 (Positive homogeneity)

If $f : \Omega \rightarrow [0, \infty]$ is measurable and $c \geq 0$, then

$$\int_{\Omega} cf \, d\mu = c \int_{\Omega} f \, d\mu. \quad (2.32)$$

Proof. This conclusion clearly holds if $c = 0$. For $c > 0$, note that $\varphi \leq cf \iff \frac{\varphi}{c} \leq f$. ■

Remark 2.7. Additivity of the integral, $\int_{\Omega} (f + g) \, d\mu = \int_{\Omega} f \, d\mu + \int_{\Omega} g \, d\mu$ for non-negative measurable f, g , is *not* immediate from the definition and requires a genuine proof; this will follow once the Monotone Convergence Theorem is available (approximate f and g by increasing sequences of simple functions, for which additivity is straightforward, and pass to the limit).

Definition 2.5 (Integral over a subset). Let $A \in \Sigma$ and let $g : \Omega \rightarrow [0, \infty]$ be measurable. The integral of g over A is defined by

$$\int_A g \, d\mu := \int_{\Omega} g \chi_A \, d\mu. \quad (2.33)$$

We now come to the first of the two fundamental convergence theorems of integration theory (the second being the Dominated Convergence Theorem, to be treated in a later lecture). Both address the same basic question: when can a limit be passed inside an integral?

Theorem 2.6 (Monotone Convergence Theorem)

Let $(f_n)_{n \geq 1}$ be a sequence of measurable functions $\Omega \rightarrow [0, \infty]$ such that

$$0 \leq f_n \leq f_{n+1} \quad \text{for every } n, \quad (2.34)$$

and let $f(x) := \lim_{n \rightarrow \infty} f_n(x)$ (which exists in $[0, \infty]$ for every x , by monotonicity). Then f is measurable, and

$$\int_{\Omega} f \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (2.35)$$

Proof. Since $(f_n(x))_n$ is non-decreasing for each x , its limit equals $\limsup_n f_n(x) = \sup_n f_n(x)$, which is measurable by .

For each n , since $f_n \leq f$, monotonicity of the integral gives

$$\int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu \quad \text{for all } n. \quad (\star)$$

The left-hand side is non-decreasing in n (again by monotonicity, since $f_n \leq f_{n+1}$) and bounded above by the fixed quantity $\int_{\Omega} f \, d\mu$, so the limit exists and

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu. \quad (2.36)$$

Fix a simple function φ with $0 \leq \varphi \leq f$, and fix a constant $\alpha \in (0, 1)$. Define

$$A_n := \{x \in \Omega : f_n(x) \geq \alpha \varphi(x)\}. \quad (2.37)$$

Each A_n is measurable (as $f_n - \alpha \varphi$ is measurable).

Claim 2.2: $A_n \subseteq A_{n+1}$ for every n .

Proof. If $f_n(x) \geq \alpha \varphi(x)$, then since $f_{n+1}(x) \geq f_n(x)$, also $f_{n+1}(x) \geq \alpha \varphi(x)$. $\square$

Claim 2.3: $\bigcup_{n=1}^{\infty} A_n = \Omega$.

Proof. Fix $x \in \Omega$. If $\varphi(x) = 0$, then $x \in A_1$ trivially (as $f_1(x) \geq 0 = \alpha\varphi(x)$). If $\varphi(x) > 0$, then

$$\alpha\varphi(x) < \varphi(x) \leq f(x). \quad (2.38)$$

Since $f_n(x) \rightarrow f(x)$, eventually $f_n(x) > \alpha\varphi(x)$, i.e. $x \in A_n$ for n large. $\square$

By monotonicity and the definition of the integral over a subset,

$$\int_{\Omega} f_n \, d\mu \geq \int_{A_n} f_n \, d\mu \geq \int_{A_n} \alpha\varphi \, d\mu = \alpha \int_{A_n} \varphi \, d\mu. \quad (2.39)$$

Since φ is simple, the set function $B \mapsto \int_B \varphi \, d\mu$ is a measure. Since (A_n) is an increasing sequence of measurable sets with $\bigcup_n A_n = \Omega$, by [Proposition 1.4](#),

$$\lim_{n \rightarrow \infty} \int_{A_n} \varphi \, d\mu = \int_{\Omega} \varphi \, d\mu. \quad (2.40)$$

Taking $n \rightarrow \infty$ in (2.39),

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \geq \alpha \int_{\Omega} \varphi \, d\mu. \quad (2.41)$$

This holds for every $\alpha \in (0, 1)$; letting $\alpha \rightarrow 1^-$ gives

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \geq \int_{\Omega} \varphi \, d\mu. \quad (2.42)$$

Finally, taking the supremum over all simple φ with $0 \leq \varphi \leq f$,

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \geq \int_{\Omega} f \, d\mu. \quad (2.43)$$

Combining both inequalities gives the result. $\blacksquare$

Remark 2.8. The restriction $\alpha < 1$ (rather than simply taking $\alpha = 1$, i.e. $A_n = \{f_n \geq \varphi\}$) is essential: without it, claim (b) above can fail. It is possible for a simple function $\varphi \leq f$ to touch f exactly at some point x , while the approximating sequence $f_n(x)$ increases to $f(x)$ without ever reaching it, so that $f_n(x) < \varphi(x)$ for every finite n .

Remark 2.9. The [Monotone Convergence Theorem](#) is, in essence, already implicit in the construction of $\int_{\Omega} f \, d\mu$. The integral of f was defined as a supremum of integrals of simple functions below f , and we exhibited an increasing sequence of simple functions converging to f everywhere. The [Monotone Convergence Theorem](#) shows that this phenomenon persists for *any* increasing sequence of non-negative measurable functions converging to f , not only sequences of simple functions.

Lemma 2.7

If $f_1, f_2 \in \mathcal{M}^+$, then $\int_{\Omega} (f_1 + f_2) d\mu = \int_{\Omega} f_1 d\mu + \int_{\Omega} f_2 d\mu$. By induction,

$$\int_{\Omega} \sum_{i=1}^n f_i d\mu = \sum_{i=1}^n \int_{\Omega} f_i d\mu. \quad (2.44)$$

Proof. By [Theorem 2.3](#), we can find increasing sequence of simple functions $\varphi_n \rightarrow f_1, \psi_n \rightarrow f_2$. Then $\varphi_n + \psi_n$ is an increasing sequence of simple function that monotonically converges to $f_1 + f_2$. By [Monotone Convergence Theorem](#),

$$\int_{\Omega} (f_1 + f_2) d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} (\varphi_n + \psi_n) d\mu \quad (2.45)$$

$$= \lim_{n \rightarrow \infty} \int_{\Omega} \varphi_n d\mu + \lim_{n \rightarrow \infty} \int_{\Omega} \psi_n d\mu \quad (2.46)$$

$$= \int_{\Omega} f_1 d\mu + \int_{\Omega} f_2 d\mu. \quad (2.47)$$

■

Proposition 2.8

If $\{f_n\}_n$ is a sequence in $\mathcal{M}^+$, then

$$\int_{\Omega} \sum_{i=1}^{\infty} f_i d\mu = \sum_{i=1}^{\infty} \int_{\Omega} f_i d\mu. \quad (2.48)$$

Proof. $g_n = \sum_{i=1}^n f_i$ is a monotone increasing sequence converging to $f = \sum_{i=1}^{\infty} f_i$. Therefore, by [Monotone Convergence Theorem](#),

$$\int_{\Omega} \sum_{i=1}^{\infty} f_i d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} g_n d\mu = \lim_{n \rightarrow \infty} \sum_{i=1}^n \int_{\Omega} f_i d\mu = \sum_{i=1}^{\infty} \int_{\Omega} f_i d\mu. \quad (2.49)$$

■

Proposition 2.9

If $f \in \mathcal{M}^+$ and $\int_{\Omega} f d\mu = 0$, then $f = 0$ μ -a.e.

Proof. Suppose $f = 0$ a.e. Let E be the set where f is nonzero, then $\mu(E) = 0$ by hypothesis. Then consider a simple function in its standard form $\varphi = \sum_{i=1}^n a_i \chi_{A_i} \leq f$, with $a_i > 0$ and A_i 's pairwise disjoint. Since $\varphi \leq f$, hence each $A_i \subseteq E$, and consequently, $\mu(A_i) = 0$. Then

$$\int_{\Omega} \varphi d\mu = \sum_{i=1}^n a_i \mu(A_i) = 0 \implies \int_{\Omega} f d\mu = \sup_{\substack{0 \leq \varphi \leq f \\ \varphi \text{ simple}}} \int_{\Omega} \varphi d\mu = 0. \quad (2.50)$$

Conversely, suppose $\int_{\Omega} f \, d\mu = 0$. Consider the set

$$E_n = \left\{ x \in \Omega : f(x) > \frac{1}{n} \right\}. \quad (2.51)$$

Then we have

$$E := \{x \in \Omega : f(x) > 0\} = \bigcup_{n=1}^{\infty} E_n. \quad (2.52)$$

We need to show that $\mu(E) = 0$. Assume for the sake of contradiction that it's strictly greater than 0. By continuity ([Proposition 1.4](#)),

$$0 < \mu(E) = \lim_{n \rightarrow \infty} \mu(E_n), \quad (2.53)$$

implies that there exists some m for which $\mu(E_m) > 0$. Then $0 \leq \frac{1}{m}\chi_{E_m} \leq f$, but then

$$\int_{\Omega} f \, d\mu \geq \int_{\Omega} \frac{1}{m}\chi_{E_m} \, d\mu = \frac{1}{m}\mu(E_m) > 0. \quad (2.54)$$

■

Theorem 2.10

If $\{f_n\}_{n \geq 1}$ and f are functions in $\mathcal{M}^+$ such that $\{f_n\}_{n \geq 1}$ converges monotonically increasing to f for μ -a.e. $x \in \Omega$. Then conclusion of [Monotone Convergence Theorem](#) still holds, i.e.

$$\int_{\Omega} f \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (2.55)$$

Proof. Suppose $f_n(x) \rightarrow f(x)$ monotonically increasing on $E \subseteq \Omega$, then $\mu(E^c) = 0$. Then $f - f\chi_E = 0$ and $f_n - f_n\chi_E = 0$ a.e. Also, $f_n\chi_E$ converges monotonically increasing to $f\chi_E$ everywhere. As a result, by [Monotone Convergence Theorem](#),

$$\int_{\Omega} f \, d\mu = \int_{\Omega} f\chi_E \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n\chi_E \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (2.56)$$

■

Proposition 2.11

If $f \in \mathcal{M}^+$, then the map

$$A \mapsto \int_A f \, d\mu = \int_{\Omega} f \cdot \chi_A \, d\mu \quad (2.57)$$

is a measure on Σ .

Proof. Suppose $\nu(A) = \int_{\Omega} f \cdot \chi_A \, d\mu$. Clearly, $\nu(A) \geq 0$. Also

$$\nu(\emptyset) = \int_{\Omega} f \cdot \chi_{\emptyset} \, d\mu = \int_{\Omega} 0 \, d\mu = 0. \quad (2.58)$$

Now, suppose $\{A_n\}_{n \geq 1}$ is a countable collection of pairwise disjoint sets. We consider

$$f_n = \sum_{i=1}^n (f \chi_{A_i}) = f \chi_{\bigcup_{i=1}^n A_i}, \quad (2.59)$$

which is a monotonically increasing sequence of measurable functions, that converge pointwise to $f \chi_{\bigcup_{i=1}^{\infty} A_i}$. Then by [Monotone Convergence Theorem](#),

$$\nu \left(\bigcup_{i=1}^{\infty} A_i \right) = \int_{\Omega} f \chi_{\bigcup_{i=1}^{\infty} A_i} d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n d\mu \quad (2.60)$$

$$= \lim_{n \rightarrow \infty} \int_{\Omega} \sum_{i=1}^n (f \chi_{A_i}) d\mu = \lim_{n \rightarrow \infty} \sum_{i=1}^n \int_{\Omega} (f \chi_{A_i}) d\mu \quad (2.61)$$

$$= \sum_{i=1}^{\infty} \nu(A_i). \quad (2.62)$$

■

§3 14 September 2026

In our last lecture, we proved that if f_n is a monotonically increasing sequence of measurable functions, then we can essentially swap limit and integral:

$$\int_{\Omega} \lim_{n \rightarrow \infty} f_n \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.1)$$

The hypothesis that (f_n) is an *increasing* sequence is essential, as the following examples show.

Example 3.1

Consider $f_n = \chi_{(n, n+1)}$. Then $f_n(x) \rightarrow 0$ pointwise for every $x \in \mathbb{R}$, yet

$$\int_{\mathbb{R}} f_n \, d\mu = 1 \quad \text{for every } n, \quad \text{while} \quad \int_{\mathbb{R}} \left(\lim_{n \rightarrow \infty} f_n \right) \, d\mu = 0. \quad (3.2)$$

So the conclusion of [Monotone Convergence Theorem](#) fails without monotonicity.

Example 3.2

Define, for $n \geq 1$,

$$f_n = n \chi_{(0, \frac{1}{n})}, \quad (3.3)$$

or, to make it continuous, a triangular spike on $[0, 1/n]$ of height $2n$ so that its graph is a triangle of area 1. Then

$$\int_0^1 f_n(x) \, dx = 1 \quad \text{for every } n, \quad (3.4)$$

but for every fixed $x > 0$ there is N such that $1/n < x$ for all $n \geq N$, so $f_n(x) \rightarrow 0$ pointwise (and $f_n(0) = 0$ for all n , by construction). Hence $f_n \rightarrow 0$ pointwise while $\int_0^1 f_n \, d\mu = 1$, again showing that monotonicity cannot be dropped.

This type of example is the classical illustration that pointwise convergence alone does not let one pass the limit inside the integral; one really needs either monotonicity or a dominating integrable function ([Dominated Convergence Theorem](#), see below).

Theorem 3.1 (Fatou's Lemma)

Let (f_n) be a sequence of nonnegative measurable functions on Ω . Then

$$\int_{\Omega} \liminf_{n \rightarrow \infty} f_n \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.5)$$

Proof. For each m , define

$$g_m(x) = \inf_{n \geq m} f_n(x). \quad (3.6)$$

For every $n \geq m$ we have $g_m \leq f_n$, and since the integral is monotone,

$$\int g_m \, d\mu \leq \int f_n \, d\mu \quad \text{for all } n \geq m. \quad (3.7)$$

Taking the infimum over $n \geq m$ on the right-hand side gives

$$\int g_m \, d\mu \leq \inf_{n \geq m} \int f_n \, d\mu. \quad (3.8)$$

The sequence (g_m) is increasing in m (an infimum over a smaller index set is larger), and

$$\lim_{m \rightarrow \infty} g_m(x) = \sup_m \inf_{n \geq m} f_n(x) = \liminf_{n \rightarrow \infty} f_n(x). \quad (3.9)$$

By the [Monotone Convergence Theorem](#) applied to (g_m) ,

$$\int \left(\liminf_{n \rightarrow \infty} f_n \right) d\mu = \lim_{m \rightarrow \infty} \int g_m \, d\mu \leq \lim_{m \rightarrow \infty} \inf_{n \geq m} \int f_n \, d\mu = \liminf_{n \rightarrow \infty} \int f_n \, d\mu. \quad (3.10)$$

■

Proposition 3.2

If $\{f_n\}_{n \geq 1}$ and f are functions in $\mathcal{M}^+$ such that $f_n(x)$ converges to $f(x)$ for μ -a.e. $x \in \Omega$, then the conclusion of [Fatou's Lemma](#) still holds, i.e.

$$\int_{\Omega} f \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.11)$$

Proof. Suppose $f_n(x) \rightarrow f(x)$ on $E \subseteq \Omega$, then $\mu(E^c) = 0$. Then $f - f\chi_E = 0$ and $f_n - f_n\chi_E = 0$ a.e. Also, $f_n\chi_E$ converges to $f\chi_E$ everywhere. As a result,

$$\int_{\Omega} f \, d\mu = \int_{\Omega} f\chi_E \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n\chi_E \, d\mu = \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.12)$$

■

Proposition 3.3

If $f \in \mathcal{M}^+$ and $\int_{\Omega} f \, d\mu < \infty$, then $\{x \in \Omega : f(x) = \infty\}$ is a null set and $\{x \in \Omega : f(x) > 0\}$ is σ -finite.

Proof. Suppose $E := \{x \in \Omega : f(x) = \infty\}$. Assume for the sake of contradiction that $\mu(E) > 0$. Then $f \geq f \cdot \chi_E$. So

$$\int_{\Omega} f \, d\mu \geq \int_{\Omega} f \cdot \chi_E \, d\mu = \int_{\Omega} \infty \cdot \chi_E \, d\mu = \infty \cdot \mu(E) = \infty, \quad (3.13)$$

which is a contradiction! Now, suppose $F = \{x \in \Omega : f(x) > 0\}$. Observe that

$$F = \bigcup_{n=1}^{\infty} \left\{ x \in \Omega : f(x) \geq \frac{1}{n} \right\} =: \bigcup_{n=1}^{\infty} F_n. \quad (3.14)$$

So it suffices to prove that each F_n has finite measure. Note that $f(x) \geq \frac{1}{n}$ on F_n , so that $n \cdot f(x) \geq 1$. In other words, $\chi_{F_n} \leq n \cdot f$. By monotonicity of integrals,

$$\mu(F_n) = \int_{\Omega} \chi_{F_n} d\mu \leq \int_{\Omega} n \cdot f d\mu = n \int_{\Omega} f d\mu < \infty. \quad (3.15)$$

■

Definition 3.1 (Positive and negative part). For a measurable function $f : \Omega \rightarrow \mathbb{R}$ define

$$f^+(x) = \max\{f(x), 0\}, \quad f^-(x) = \max\{-f(x), 0\}, \quad (3.16)$$

to be the **positive part** and **negative part** of f , respectively.

Remark 3.1. It's clear that $f^+, f^- \geq 0$ are measurable, and

$$f = f^+ - f^-, \quad |f| = f^+ + f^-. \quad (3.17)$$

This decomposition is not unique as a way of writing f as a difference of two nonnegative functions (one may add the same nonnegative function to both f^+ and f^-), but it is the canonical choice, and it is the one used to define the integral of a general function.

Definition 3.2 (The space L^1). Let (Ω, μ) be a measure space. Define

$$L^1(\Omega, \mu) = \left\{ f : \Omega \rightarrow \mathbb{R} \text{ measurable} : \int_{\Omega} |f| d\mu < \infty \right\}. \quad (3.18)$$

For $f \in L^1(\Omega, \mu)$, both $\int f^+ d\mu$ and $\int f^- d\mu$ are finite (since their sum is $\int |f| d\mu < \infty$), and we define

$$\int_{\Omega} f d\mu := \int_{\Omega} f^+ d\mu - \int_{\Omega} f^- d\mu. \quad (3.19)$$

Remark 3.2. The integral is first defined for simple functions, then extended to nonnegative measurable functions via the supremum definition, and finally extended to L^1 via this difference, precisely so as to avoid ever having to make sense of $\infty - \infty$.

Proposition 3.4

If $f \in L^1$, then

$$\left| \int_{\Omega} f d\mu \right| \leq \int_{\Omega} |f| d\mu. \quad (3.20)$$

Proof.

$$\left| \int_{\Omega} f \, d\mu \right| = \left| \int_{\Omega} f^+ \, d\mu - \int_{\Omega} f^- \, d\mu \right| \leq \int_{\Omega} f^+ \, d\mu + \int_{\Omega} f^- \, d\mu = \int_{\Omega} |f| \, d\mu. \quad (3.21)$$

■

Proposition 3.5

If $f, g \in L^1(\Omega, \mu)$ and $c \in \mathbb{R}$, then $f + cg \in L^1(\Omega, \mu)$ and

$$\int_{\Omega} (f + cg) \, d\mu = \int_{\Omega} f \, d\mu + c \int_{\Omega} g \, d\mu. \quad (3.22)$$

Proof. First notice that

$$|f + cg| \leq |f| + |c| |g|, \quad (3.23)$$

which implies using monotonicity of integral of non-negative functions that

$$\int_{\Omega} |f + cg| \, d\mu \leq \int_{\Omega} |f| + |c| |g| \, d\mu = \int_{\Omega} |f| \, d\mu + |c| \int_{\Omega} |g| \, d\mu < \infty. \quad (3.24)$$

We'll now show that $\int_{\Omega} cf \, d\mu = c \int_{\Omega} f \, d\mu$. For $c \geq 0$,

$$(cf)^+ = cf^+, (cf)^- = cf^-. \quad (3.25)$$

Then

$$\int_{\Omega} cf \, d\mu = \int_{\Omega} (cf)^+ \, d\mu - \int_{\Omega} (cf)^- \, d\mu = \int_{\Omega} cf^+ \, d\mu - \int_{\Omega} cf^- \, d\mu = c \int_{\Omega} f \, d\mu. \quad (3.26)$$

For $c < 0$,

$$(cf)^+ = -cf^-, (cf)^- = -cf^+. \quad (3.27)$$

Then

$$\int_{\Omega} cf \, d\mu = \int_{\Omega} (cf)^+ \, d\mu - \int_{\Omega} (cf)^- \, d\mu = \int_{\Omega} -cf^- \, d\mu - \int_{\Omega} -cf^+ \, d\mu = c \int_{\Omega} f \, d\mu. \quad (3.28)$$

Now let's show that $\int_{\Omega} f + g \, d\mu = \int_{\Omega} f \, d\mu + \int_{\Omega} g \, d\mu$. Suppose $h = f + g$. Then

$$h^+ - h^- = f^+ - f^- + g^+ - g^- \implies h^+ + f^- + g^- = h^- + f^+ + g^+. \quad (3.29)$$

Using the additivity of integral of non-negative functions, we get

$$\int_{\Omega} h^+ \, d\mu + \int_{\Omega} f^- \, d\mu + \int_{\Omega} g^- \, d\mu = \int_{\Omega} h^- \, d\mu + \int_{\Omega} f^+ \, d\mu + \int_{\Omega} g^+ \, d\mu. \quad (3.30)$$

Regrouping gives us the desired result:

$$\int_{\Omega} h^+ \, d\mu - \int_{\Omega} h^- \, d\mu = + \int_{\Omega} f^+ \, d\mu - \int_{\Omega} f^- \, d\mu + \int_{\Omega} g^+ \, d\mu - \int_{\Omega} g^- \, d\mu, \quad (3.31)$$

i.e. $\int_{\Omega} h \, d\mu = \int_{\Omega} f \, d\mu + \int_{\Omega} g \, d\mu$. ■

Lemma 3.6

If $f \leq g$ a.e. then $\int_{\Omega} f \, d\mu \leq \int_{\Omega} g \, d\mu$.

Proof. Suppose $f \leq g$ on E with $\mu(E^c) = 0$. Then $(g - f)\chi_E \geq 0$. The integral of a non-negative function is non-negative, so

$$\int_{\Omega} g \, d\mu - \int_{\Omega} f \, d\mu = \int_{\Omega} (g - f) \, d\mu = \int_{\Omega} (g - f)\chi_E \, d\mu \geq 0. \quad (3.32)$$
■

Proposition 3.7

If $f \in M$ and $g \in L^1(\Omega, \mu)$ with $|f| \leq g$ a.e., then $f \in L^1(\Omega, \mu)$. (A function dominated by an integrable function is itself integrable.)

Proof. By monotonicity of integrals,

$$\int_{\Omega} |f| \, d\mu \leq \int_{\Omega} g \, d\mu < \infty. \quad (3.33)$$
■

Proposition 3.8

If $f, g \in L^1$, then the following are equivalent:

- (a) $\int_A f \, d\mu = \int_A g \, d\mu$ for every $A \in \Sigma$;
- (b) $\int_{\Omega} |f - g| \, d\mu = 0$;
- (c) $f = g$ a.e.

Proof. The equivalence of (b) and (c) follows from [Proposition 2.9](#):

$$\int_{\Omega} |f - g| \, d\mu = 0 \iff |f - g| = 0 \text{ a.e.} \iff f = g \text{ a.e.} \quad (3.34)$$

(b) $\Rightarrow$ (a) Take any $A \in \Sigma$. Then

$$\left| \int_A f \, d\mu - \int_A g \, d\mu \right| = \left| \int_{\Omega} (f - g)\chi_A \, d\mu \right| \leq \int_{\Omega} |f - g|\chi_A \, d\mu \leq \int_{\Omega} |f - g| \, d\mu = 0. \quad (3.35)$$

(a) $\Rightarrow$ (c) Suppose $A = \{x \in \Omega : f(x) > g(x)\}$, which is a measurable set. On A , $f - g > 0$. If $\mu(A) > 0$, then $\int_A (f - g) > 0$ by the following claim, which contradicts with the hypothesis.

Claim 3.1: If $\mu(E) > 0$ and $h > 0$ on E , then $\int_E h \, d\mu > 0$.

Proof. Let $E_n = \{x \in E : h(x) \geq \frac{1}{n}\}$. Then $\{E_n\}_n$ is an increasing sequence with

$$E = \bigcup_{i=1}^{\infty} E_n. \quad (3.36)$$

By continuity (Proposition 1.4),

$$0 < \mu(E) = \lim_{n \rightarrow \infty} \mu(E_n), \quad (3.37)$$

implies that there exists some m for which $\mu(E_m) > 0$. Then $0 \leq \frac{1}{m} \chi_{E_m} \leq h$, but then

$$\int_E h \, d\mu \geq \int_{\Omega} \frac{1}{m} \chi_{E_m} \, d\mu = \frac{1}{m} \mu(E_m) > 0. \quad (3.38)$$

□

Therefore, $\mu(A) = 0$. Similarly, if $B = \{x \in \Omega : f(x) < g(x)\}$, then $\mu(B) = 0$. Now,

$$\{x \in \Omega : f(x) \neq g(x)\} = A \cup B, \quad (3.39)$$

is also a null set, since it's the union of two null sets. ■

Remark 3.3. Note that $\int_{\Omega} f \, d\mu = \int_{\Omega} g \, d\mu$ does not necessarily imply that $f = g$ a.e. For example, consider $\Omega = [0, 2\pi]$ and $f(x) = \sin x$, $g(x) = \cos x$. Then

$$\int_0^{2\pi} \sin x \, dx = \int_0^{2\pi} \cos x \, dx = 0, \quad (3.40)$$

but it's **NOT** true that $\sin x = \cos x$ a.e.

Theorem 3.9 (Dominated Convergence Theorem)

Let (f_n) be a sequence of measurable functions on Ω such that $f_n \rightarrow f$ pointwise a.e., and suppose there is a function $g \in L^1(\Omega, \mu)$ such that

$$|f_n(x)| \leq g(x) \text{ a.e. for all } n. \quad (3.41)$$

Then $f \in L^1(\Omega, \mu)$ and

$$\int_{\Omega} f \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.42)$$

Proof. Since $|f_n| \leq g$ for every n and $f_n \rightarrow f$ pointwise, also $|f| \leq g$, so $f \in L^1(\Omega, \mu)$ Proposition 3.7.

Since $g + f_n \geq 0$ (because $|f_n| \leq g$) and $g + f_n \rightarrow g + f$ pointwise a.e., [Fatou's Lemma](#) gives

$$\int_{\Omega} (g + f) \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} (g + f_n) \, d\mu = \int_{\Omega} g \, d\mu + \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.43)$$

Since $\int_{\Omega} g \, d\mu$ is finite, it may be cancelled from both sides:

$$\int_{\Omega} f \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \quad (3.44)$$

Likewise $g - f_n \geq 0$ and $g - f_n \rightarrow g - f$ pointwise a.e., so [Fatou's Lemma](#) gives

$$\int_{\Omega} (g - f) \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} (g - f_n) \, d\mu = \int_{\Omega} g \, d\mu - \limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu, \quad (3.45)$$

using $\liminf(-a_n) = -\limsup(a_n)$. Cancelling $\int_{\Omega} g \, d\mu$ (finite) from both sides:

$$\limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu. \quad (3.46)$$

Combining (3.44) and (3.46),

$$\limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu, \quad (3.47)$$

and since always $\liminf \leq \limsup$, all three quantities coincide. Hence $\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu$ exists and equals $\int_{\Omega} f \, d\mu$. ■

Remark 3.4. The [Monotone Convergence Theorem](#), [Fatou's Lemma](#), and the [Dominated Convergence Theorem](#) are the three fundamental convergence theorems needed to justify moving a limit inside an integral: either the sequence is monotone (Monotone Convergence), or it is dominated by a single integrable function (Dominated Convergence); [Fatou's Lemma](#) always applies to nonnegative sequences and is the tool used to prove the other two.

Theorem 3.10 (Generalized Dominated Convergence Theorem)

If $f_n, g_n, g \in L^1$, $f_n \rightarrow f$ and $g_n \rightarrow g$ a.e.,

$$|f_n| \leq g_n, \quad \text{and} \quad \lim_{n \rightarrow \infty} \int_{\Omega} g_n \, d\mu = \int_{\Omega} g \, d\mu, \quad (3.48)$$

then $f \in L^1$, and

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu = \int_{\Omega} f \, d\mu. \quad (3.49)$$

Proof. Since $|f_n| \leq g_n$ for every n and $f_n \rightarrow f$ pointwise a.e. and $g_n \rightarrow g$ a.e., we have that $|f| \leq g$ a.e. So $f \in L^1(\Omega, \mu)$ [Proposition 3.7](#).

Since $g_n + f_n \geq 0$ (because $|f_n| \leq g_n$) and $g_n + f_n \rightarrow g + f$ pointwise a.e., [Fatou's Lemma](#) gives

$$\begin{aligned}
\int_{\Omega} g \, d\mu + \int_{\Omega} f \, d\mu &= \int_{\Omega} (g + f) \, d\mu \\
&\leq \liminf_{n \rightarrow \infty} \int_{\Omega} (g_n + f_n) \, d\mu \\
&= \lim_{n \rightarrow \infty} \int_{\Omega} g_n \, d\mu + \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \\
&= \int_{\Omega} g \, d\mu + \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu.
\end{aligned} \tag{3.50}$$

Since $\int_{\Omega} g \, d\mu$ is finite, it may be cancelled from both sides:

$$\int_{\Omega} f \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu. \tag{3.51}$$

Likewise $g_n - f_n \geq 0$ and $g_n - f_n \rightarrow g - f$ pointwise a.e., so [Fatou's Lemma](#) gives

$$\begin{aligned}
\int_{\Omega} g \, d\mu - \int_{\Omega} f \, d\mu &= \int_{\Omega} (g - f) \, d\mu \\
&\leq \liminf_{n \rightarrow \infty} \int_{\Omega} (g_n - f_n) \, d\mu \\
&= \lim_{n \rightarrow \infty} \int_{\Omega} g_n \, d\mu - \limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \\
&= \int_{\Omega} g \, d\mu - \limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu.
\end{aligned} \tag{3.52}$$

Cancelling $\int_{\Omega} g \, d\mu$ (finite) from both sides:

$$\limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu. \tag{3.53}$$

Combining (3.51) and (3.53),

$$\limsup_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu, \tag{3.54}$$

and since always $\liminf \leq \limsup$, all three quantities coincide. Hence $\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu$ exists and equals $\int_{\Omega} f \, d\mu$. ■

Remark 3.5. In the proof of [Generalized Dominated Convergence Theorem](#), we used the fact that if a_n, b_n are two sequences such that $a_n \rightarrow a$, then

$$\liminf_{n \rightarrow \infty} (a_n + b_n) = a + \liminf_{n \rightarrow \infty} b_n, \tag{3.55}$$

$$\liminf_{n \rightarrow \infty} (a_n - b_n) = a - \limsup_{n \rightarrow \infty} b_n. \tag{3.56}$$

To prove the first one, notice that

$$\inf_{m \geq n} (a_m + b_m) \geq \inf_{m \geq n} a_m + \inf_{m \geq n} b_m. \tag{3.57}$$

Taking $n \rightarrow \infty$, we get

$$\liminf_{n \rightarrow \infty} (a_n + b_n) \geq \liminf_{n \rightarrow \infty} a_n + \liminf_{n \rightarrow \infty} b_n \quad (3.58)$$

$$= a + \liminf_{n \rightarrow \infty} b_n. \quad (3.59)$$

On the other hand, writing $b_n = (a_n + b_n) + (-a_n)$ and applying (3.58), we get

$$\liminf_{n \rightarrow \infty} b_n = \liminf_{n \rightarrow \infty} [(a_n + b_n) + (-a_n)] \quad (3.60)$$

$$\geq \liminf_{n \rightarrow \infty} (a_n + b_n) + \liminf_{n \rightarrow \infty} (-a_n) \quad (3.61)$$

$$= \liminf_{n \rightarrow \infty} (a_n + b_n) - a. \quad (3.62)$$

Therefore, combining (3.59) and (3.62), we have established (3.55). To see (3.56),

$$\liminf_{n \rightarrow \infty} (a_n - b_n) = a + \liminf_{n \rightarrow \infty} (-b_n) = a - \limsup_{n \rightarrow \infty} b_n.$$

Example 3.3

Let

$$f_n(x) = n \chi_{(0,1/n)}(x), \quad x \in [0, 1]. \quad (3.63)$$

Then $f_n(x) \rightarrow 0$ for every $x \in [0, 1]$ (for $x = 0$ every $f_n(0) = 0$; for $x > 0$, eventually $\frac{1}{n} < x$), so $f_n \rightarrow 0$ pointwise. However,

$$\int_0^1 f_n dx = 1 \quad \text{for every } n, \quad \text{while } \int_0^1 \left(\lim_{n \rightarrow \infty} f_n \right) dx = 0. \quad (3.64)$$

The conclusion of the Dominated Convergence Theorem fails here because no single integrable function g dominates all the f_n . The pointwise supremum of the family is essentially $g(x) = \frac{1}{x}$, and

$$\int_0^1 \frac{1}{x} dx = \infty, \quad (3.65)$$

so $g \notin L^1([0, 1])$. This also shows the Monotone Convergence Theorem does not apply, since the sequence (f_n) is not pointwise increasing.

We now show that one cannot hope to define a measure on *all* subsets of $\mathbb{R}$ that

- (i) is translation invariant,
- (ii) is monotone (if $A \subseteq B$ then $\mu(A) \leq \mu(B)$), and
- (iii) and that extends the usual length of intervals.

Consequently the σ -algebra of measurable sets must be a strict subset of the power set $\mathcal{P}(\mathbb{R})$. For this, we illustrate the following example:

On $[0, 1]$, declare $x \sim y$ if $x - y \in \mathbb{Q}$. This is an equivalence relation, partitioning $[0, 1]$ into equivalence classes. By the Axiom of Choice, choose exactly one representative x_I from each

equivalence class I , and let

$$A = \{x_I : I \in [0, 1] / \sim\} \subseteq [0, 1], \quad (3.66)$$

where I indexes the set of equivalence classes. The set A is called a *Vitali set*.

Theorem 3.11

There is no measure μ on $\mathcal{P}(\mathbb{R})$ that is translation invariant, monotone, and assigns to every interval its usual length.

Proof. Suppose such a μ exists. For $r \in \mathbb{Q} \cap [-1, 1]$ set

$$Y_r = A + r = \{a + r : a \in A\}. \quad (3.67)$$

If $Y_r \cap Y_s \neq \emptyset$ for $r \neq s \in \mathbb{Q}$, then $x_i + r = x_j + s$ for some representatives, so

$$x_i - x_j = s - r \in \mathbb{Q}, \quad (3.68)$$

forcing $x_i \sim x_j$; since A contains only one representative per class, $x_i = x_j$, hence $r = s$, a contradiction. So the sets $\{Y_r\}_{r \in \mathbb{Q} \cap [-1, 1]}$ are pairwise disjoint.

Since $A \subseteq [0, 1]$ and $r \in [-1, 1]$,

$$Y_r \subseteq [-1, 2] \quad \text{for every } r \in \mathbb{Q} \cap [-1, 1]. \quad (3.69)$$

For any $x \in [0, 1]$ there is a representative $x_i \sim x$ with $x - x_i \in \mathbb{Q} \cap [-1, 1]$, so $x \in Y_{x-x_i}$. Hence

$$[0, 1] \subseteq \bigcup_{r \in \mathbb{Q} \cap [-1, 1]} Y_r \subseteq [-1, 2]. \quad (3.70)$$

By translation invariance, $\mu(Y_r) = \mu(A)$ for every r . By monotonicity and countable additivity on the disjoint countable union,

$$\mu([0, 1]) \leq \mu\left(\bigcup_{r \in \mathbb{Q} \cap [-1, 1]} Y_r\right) = \sum_{r \in \mathbb{Q} \cap [-1, 1]} \mu(Y_r) = \sum_{r \in \mathbb{Q} \cap [-1, 1]} \mu(A) \leq \mu([-1, 2]). \quad (3.71)$$

Now $\mu([0, 1]) = 1$ and $\mu([-1, 2]) = 3$ (extending the usual length), so

$$1 \leq \sum_{r \in \mathbb{Q} \cap [-1, 1]} \mu(A) \leq 3. \quad (3.72)$$

But $\mathbb{Q} \cap [-1, 1]$ is countably infinite, so the middle sum is a countable sum of a single fixed value $\mu(A) \in [0, \infty]$:

- if $\mu(A) = 0$, the sum is 0, contradicting $1 \leq 0$;
- if $\mu(A) > 0$, the sum is $+\infty$, contradicting $\infty \leq 3$.

Either way we reach a contradiction. Hence no such measure μ can exist on all of $\mathcal{P}(\mathbb{R})$. ■

Remark 3.6. This is the price one pays for wanting a translation-invariant, monotone measure extending length: it cannot be defined on *every* subset of $\mathbb{R}$. This motivates restricting attention to a suitable σ -algebra (the Lebesgue measurable sets), which is the object to be constructed next.

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Having seen that no measure on *all* subsets of $\mathbb{R}$ can extend length while remaining translation invariant, we now begin the actual construction of Lebesgue measure. The construction proceeds geometrically, starting from rectangles (objects simple enough that their “volume” is unambiguous) and builds up to general open, and eventually general measurable, sets. This lecture develops the purely geometric facts about rectangles and open sets that the construction will rely on.

Definition 4.1 (Rectangle). A (closed) **rectangle** $R \subseteq \mathbb{R}^d$ is a product of d closed, bounded intervals:

$$R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_d, b_d], \quad a_i \leq b_i. \quad (4.1)$$

The sides of R are parallel to the coordinate axes; rotated rectangles are not considered. The associated **open rectangle** is

$$R^\circ := (a_1, b_1) \times (a_2, b_2) \times \cdots \times (a_d, b_d). \quad (4.2)$$

Definition 4.2 (Volume). The **volume** of the rectangle $R = \prod_{i=1}^d [a_i, b_i]$ is

$$|R| := \prod_{i=1}^d (b_i - a_i). \quad (4.3)$$

(The same formula, with the same value, defines the volume $|R^\circ|$ of the associated open rectangle.)

Definition 4.3 (Almost disjoint collection). A finite (or countable) collection of rectangles $\{R_k\}$ is **almost disjoint** if the open rectangles R_k° are pairwise disjoint; that is, the rectangles may touch along their boundaries (share a face, edge, or corner), but their interiors do not overlap.

Proposition 4.1

Suppose a rectangle R is the almost disjoint union of finitely many rectangles $R_1, \dots, R_n$, i.e. $R = \bigcup_{k=1}^n R_k$ with $\{R_k\}$ almost disjoint. Then

$$|R| = \sum_{k=1}^n |R_k|. \quad (4.4)$$

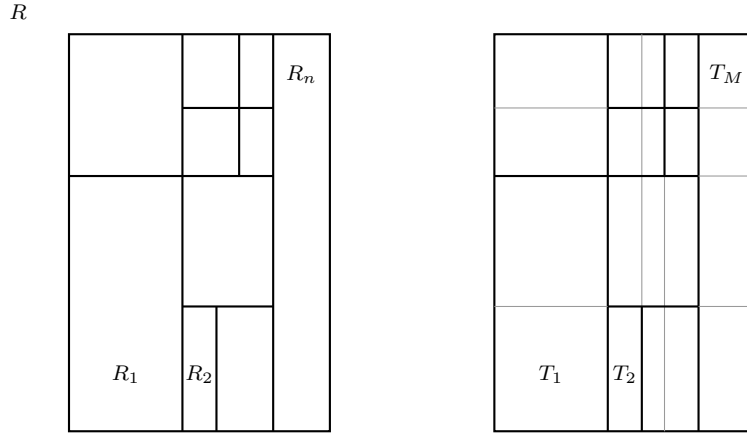
Proof. Case 1: Suppose first that each side of $R = \prod_{i=1}^d [a_i, b_i]$ has been partitioned into finitely many subintervals. Say the i -th side $[a_i, b_i]$ is cut into pieces of lengths $\ell_{i,1}, \dots, \ell_{i,m_i}$ with $\sum_j \ell_{i,j} = b_i - a_i$, and the R_k are exactly the resulting grid cells (one for each choice of a

subinterval in each coordinate). Then, using distributivity of multiplication over addition in each coordinate,

$$|R| = \prod_{i=1}^d (b_i - a_i) = \prod_{i=1}^d \left(\sum_{j=1}^{m_i} \ell_{i,j} \right) = \sum_{j_1=1}^{m_1} \cdots \sum_{j_d=1}^{m_d} \ell_{1,j_1} \cdots \ell_{d,j_d} = \sum_k |R_k|, \quad (4.5)$$

since each term in the expanded product is exactly the volume of one grid cell, and every grid cell arises exactly once.

Case 2: Given an arbitrary almost disjoint decomposition $R = \bigcup_{k=1}^n R_k$, extend every face (bounding hyperplane) of every R_k across the entirety of R , in every coordinate direction.



This produces a common refining grid on R : each original rectangle R_k is itself partitioned by this grid into finitely many grid cells (an almost disjoint union of sub-rectangles), and by Case 1 applied to R_k ,

$$|R_k| = \sum_{\substack{\text{grid cells } T \\ T \subseteq R_k}} |T|. \quad (4.6)$$

On the other hand, since $\{R_k\}$ is almost disjoint and covers R , every grid cell T of the common refinement lies in exactly one R_k (up to boundary overlaps, which do not affect volume), so summing over k ,

$$\sum_{k=1}^n |R_k| = \sum_{k=1}^n \sum_{T \subseteq R_k} |T| = \sum_{\text{all grid cells } T \text{ of } R} |T| = |R|, \quad (4.7)$$

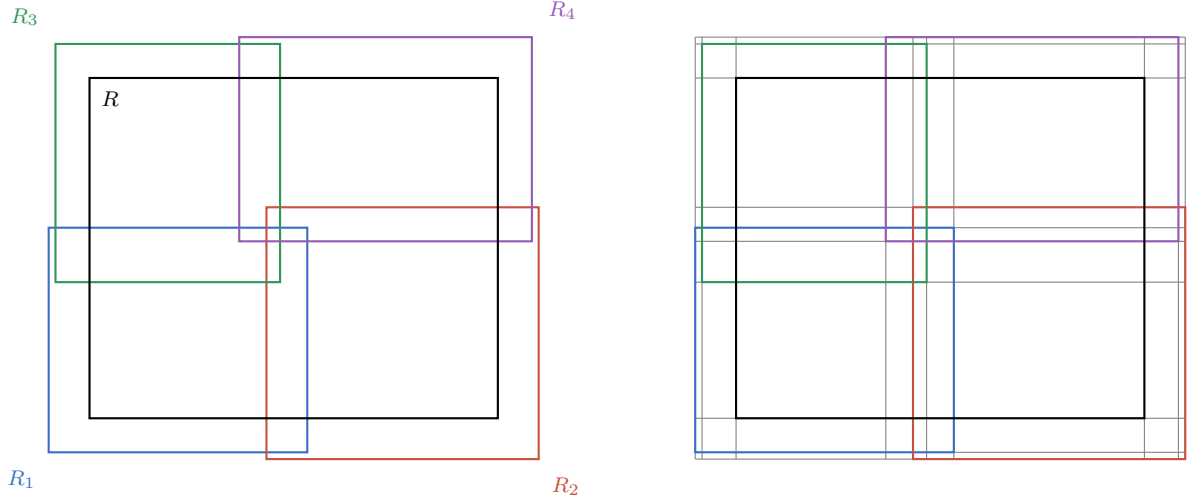
the last equality again by Case 1, applied this time to R itself with the common refining grid. ■

Proposition 4.2

Suppose a rectangle R is covered by finitely many rectangles $R_1, \dots, R_n$ (i.e. $R \subseteq \bigcup_{k=1}^n R_k$), with no assumption that the R_k are disjoint or almost disjoint, and no assumption that $R_k \subseteq R$. Then

$$|R| \leq \sum_{k=1}^n |R_k|. \quad (4.8)$$

Proof. As in the proof of [Proposition 4.1](#), extend every face of every R_k and R to form a refining grid. This may create grid rectangles that are **NOT** in any of the R_i 's or R .



This decomposes R into finitely many almost disjoint grid cells $T_1, \dots, T_m$, so that $R = \bigcup_j T_j$ almost disjointly, and by [Proposition 4.1](#),

$$|R| = \sum_{j=1}^m |T_j|. \quad (4.9)$$

Since $R \subseteq \bigcup_k R_k$, every grid cell T_j lies in at least one R_k ; assign each T_j to one such R_k (breaking ties arbitrarily).

$$|R| = \sum_{j=1}^m |T_j| = \sum_{k=1}^n \left(\sum_{\substack{T_j \text{ assigned} \\ \text{to } R_k}} |T_j| \right). \quad (4.10)$$

For a fixed k , the cells T_j assigned to R_k are almost disjoint and all contained in $R_k \cap R \subseteq R_k$. Note that they may not exhaust all of R_k . In fact, they are a subset of all the grid rectangles that are contained in R_k . Therefore,

$$\sum_{\substack{T_j \text{ assigned} \\ \text{to } R_k}} |T_j| \leq \sum_{\substack{T \text{ grid rectangle,} \\ T \subseteq R_k}} |T| = |R_k|, \quad (4.11)$$

where the last equality follows from [Proposition 4.1](#). Now combining (4.10) and (4.11), we are done! ■

Remark 4.1. [Proposition 4.2](#) is the geometric heart of **countable subadditivity** for the outer measure to be constructed in the next lecture: it says that volume cannot be “cheated” by covering a rectangle with other rectangles of smaller total volume.

Proposition 4.3 (Structure of open subsets of $\mathbb{R}$)

Every open set $O \subseteq \mathbb{R}$ can be written as a countable union of pairwise disjoint open intervals. Moreover, this decomposition is unique: it is exactly the decomposition of O into its connected components.

Proof. For $x \in O$, let

$$I_x := (a_x, b_x), \text{ where } a_x = \inf \{a : (a, x) \subseteq O\}, \text{ and } b_x = \sup \{b : (x, b) \subseteq O\}, \quad (4.12)$$

be the *largest* open interval contained in O that contains x . Since O is open, I_x is nonempty (it contains some small interval around x), and clearly $O = \bigcup_{x \in O} I_x$.

If $x, y \in O$ and $I_x \cap I_y \neq \emptyset$, then $I_x \cup I_y$ is itself an open interval (a union of two overlapping intervals) contained in O and containing both x and y ; by maximality of I_x (as the *largest* such interval containing x), we must have

$$I_x \cup I_y \subseteq I_x \implies I_y \subseteq I_x. \quad (4.13)$$

Similarly, $I_x \subseteq I_y$, so $I_x = I_y$. Hence, the collection $\{I_x : x \in O\}$ consists of intervals that are pairwise either identical or disjoint; selecting one representative from each distinct interval gives a collection of pairwise disjoint open intervals $\{I_j\}_{j \in J}$ with $O = \bigsqcup_{j \in J} I_j$.

Each I_j is a nonempty open interval, hence contains a rational number q_j ; since the I_j are pairwise disjoint, distinct intervals contain distinct rationals, giving an injection

$$\begin{aligned} J &\rightarrow \mathbb{Q} \\ j &\mapsto q_j. \end{aligned} \quad (4.14)$$

An injection into a countable set shows the index set J is countable.

Now about the uniqueness, suppose

$$O = \bigsqcup_k J_k, \quad (4.15)$$

is another decomposition into pairwise disjoint open intervals. Each J_k , being a connected subset of O , is contained in the maximal interval through any of its points, i.e. $J_k \subseteq I_j$ for a **unique** j .

Conversely, I_j is covered by those J_k 's contained in it. But I_j is connected and the J_k 's are open and pairwise disjoint, so I_j cannot be covered by two or more nonempty disjoint open pieces (else I_j would be disconnected).

Hence exactly one J_k satisfies $J_k \subseteq I_j$ with the covering forcing $J_k = I_j$. This gives a bijection between the two decompositions, proving uniqueness. ■

Proposition 4.3 is special to $\mathbb{R}$: it relies essentially on the fact that connected open subsets of $\mathbb{R}$ are exactly the open intervals. In $\mathbb{R}^d$ for $d \geq 2$, connected open sets can have arbitrarily complicated shapes (e.g. an open disk), so an open set in $\mathbb{R}^d$ cannot in general be written as a

disjoint union of open *rectangles*. The right substitute uses **closed rectangles**, allowing *almost* disjointness in place of disjointness.

Remark 4.2. In $d = 1$, consider the open set $(0, 1) \subseteq \mathbb{R}$. Then we can write $(0, 1)$ as a countable union of almost-disjoint closed intervals:

$$(0, 1) = \bigcup_{n=2}^{\infty} \left[\frac{1}{n+1}, \frac{1}{n} \right] \cup \bigcup_{n=2}^{\infty} \left[1 - \frac{1}{n}, 1 - \frac{1}{n+1} \right]. \quad (4.16)$$

Definition 4.4 (Dyadic cubes). For $n \in \mathbb{Z}$, a **dyadic cube of generation n** is a set of the form

$$Q = \prod_{i=1}^d \left[\frac{k_i}{2^n}, \frac{k_i+1}{2^n} \right], \quad k_1, \dots, k_d \in \mathbb{Z}. \quad (4.17)$$

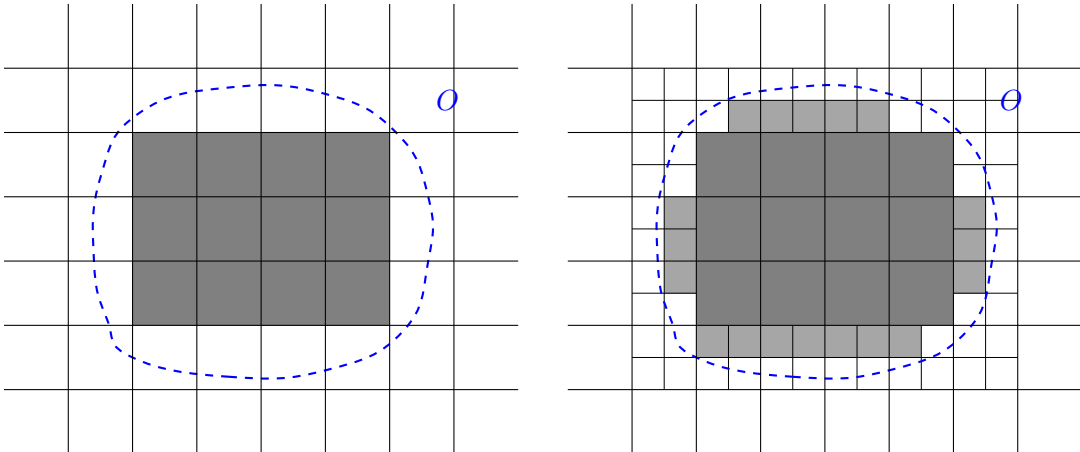
For fixed n , the dyadic cubes of generation n have disjoint interiors and their union is all of $\mathbb{R}^d$ (they form a grid of cubes of side length 2^{-n}). Every dyadic cube Q of generation n is contained in a unique dyadic cube $\tilde{Q}$ of generation $n-1$ (its **parent**, of side length $2^{-(n-1)}$), obtained by halving the grid spacing; conversely, each dyadic cube of generation $n-1$ is the almost disjoint union of exactly 2^d dyadic cubes of generation n (its **children**).

Remark 4.3. Two dyadic cubes (of possibly different generations) are always either almost disjoint (disjoint interiors) or one is contained in the other. This nesting property is the key structural fact that makes the dyadic grid useful.

Proposition 4.4 (Structure of open subsets of $\mathbb{R}^d$)

Every open set $O \subseteq \mathbb{R}^d$ can be written as a countable, almost disjoint union of closed rectangles.

Proof. Call a dyadic cube Q **selected** if $Q \subseteq O$ but its parent $\tilde{Q}$ is *not* contained in O (i.e. Q is a maximal dyadic cube contained in O , in the sense that enlarging it to its parent would exit O).



Define

$$\mathcal{Q} := \{Q : Q \text{ is a selected dyadic cube}\}, \quad U := \bigcup_{Q \in \mathcal{Q}} Q. \quad (4.18)$$

We claim $O = U$. $U \subseteq O$ is immediate, since every selected Q satisfies $Q \subseteq O$ by definition.

Let $x \in O$. Since O is open, there is $\varepsilon > 0$ with the open ball $B(x, \varepsilon) \subseteq O$. Choose n large enough that a dyadic cube of generation n has diameter

$$2^{-n}\sqrt{d} < \varepsilon. \quad (4.19)$$

Then there is a generation- n dyadic cube Q_n containing x that satisfies

$$Q_n \subseteq B(x, \varepsilon) \subseteq O. \quad (4.20)$$

Since $Q_n \subseteq O$, that means Q_n has been selected at step n , or one of its previous generation parents has been selected at an earlier step. Therefore, $x \in U = \bigcup_{Q \in \mathcal{Q}} Q$.

Now we show that $\mathcal{Q}$ is an almost-disjoint collection. Suppose $Q, Q' \in \mathcal{Q}$ are distinct selected cubes with overlapping interiors. By the nesting property, one is contained in the other; say

$$Q' \subsetneq Q, \quad (4.21)$$

so that Q' is a proper descendant of Q , of strictly higher generation. Let Q'' be the immediate parent of Q' . Then

$$Q' \subseteq Q'' \subseteq Q. \quad (4.22)$$

Since $Q \subseteq O$ (as Q is selected), also $Q'' \subseteq O$. But this contradicts Q' being selected, which requires the immediate parent of Q' to *not* be contained in O . Hence no such containment can occur, and the interiors of distinct selected cubes are disjoint.

Finally, we'll now show that $\mathcal{Q}$ is a countable collection. For each fixed generation $n \in \mathbb{Z}$, the dyadic cubes of generation n are indexed by $\mathbb{Z}^d$, a countable set.

$$\mathcal{Q} \subseteq \bigcup_{n \in \mathbb{Z}} \{\text{generation-}n \text{ dyadic cubes}\}, \quad (4.23)$$

which is a countable union of countable sets, hence countable. ■

Remark 4.4 (Comparison with the one-dimensional case). For $d = 1$, the construction in [Proposition 4.4](#) produces a decomposition into closed dyadic *intervals*, which is generally coarser and less canonical than the connected-components decomposition of [Proposition 4.3](#) (dyadic intervals need not align with the true connected components of O , and distinct choices of the underlying grid (e.g. shifting or rescaling it) give different, equally valid, decompositions). This is consistent with the failure of *uniqueness* in higher dimensions: [Proposition 4.4](#) asserts existence of an almost disjoint decomposition into closed rectangle, not that it is the unique or canonical one.

Remark 4.5. The decomposition of [Proposition 4.4](#) cannot in general be made into a genuinely disjoint union of *closed* cubes (closed cubes covering a connected open set must share boundary faces where they meet), nor into a disjoint union of *open* rectangles when $d \geq 2$ (e.g. an open disk is not a union of finitely or countably many disjoint open rectangles, since removing the boundary of any inscribed rectangle from a disk leaves a connected leftover region that cannot itself be a union of open rectangles disjoint from the first without gaps). Allowing cubes to touch along boundaries (“almost” disjoint) is what makes the construction possible in every dimension.

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We now begin the construction proper. The strategy has two stages: first, define a set function m^* , the **exterior measure** (or **outer measure**), on *every* subset of $\mathbb{R}^d$ by covering from outside with cubes; second, single out a sub-collection of “good” sets (we’ll call them the **Lebesgue measurable sets**) on which m^* will turn out to be countably additive, i.e. a genuine measure in the sense of Section 3. [Theorem 3.11](#) tells us this sub-collection cannot be all of $\mathcal{P}(\mathbb{R}^d)$.

Definition 5.1 (Exterior measure). For a set $E \subseteq \mathbb{R}^d$, the **exterior measure** of E is

$$m^*(E) := \inf \left\{ \sum_{j=1}^{\infty} |Q_j| : E \subseteq \bigcup_{j=1}^{\infty} Q_j, \quad Q_j \text{ closed cubes} \right\} \in [0, \infty]. \quad (5.1)$$

That is, $m^*(E)$ is the infimum, over all countable covers of E by closed cubes (not required to be disjoint or almost disjoint), of the total volume of the cubes used.

Remark 5.1. Every set $E \subseteq \mathbb{R}^d$ admits at least one such cover (e.g. by cubes tiling all of $\mathbb{R}^d$), so $m^*(E)$ is always a well-defined element of $[0, \infty]$.

Remark 5.2 (Why cover from outside). One might instead try to define an “interior” measure by taking the *supremum* of the total volume of cubes contained *inside* E . This alternative fails badly. For example, a set with empty interior but which is, intuitively, “large” (such as the set of irrational numbers in $[0, 1]$, or more strikingly a dense set with no interior points at all) would have interior measure 0, since no cube of positive volume can fit inside it. Covering from *outside* does not suffer from this defect, which is why the construction proceeds this way.

Remark 5.3 (Cubes versus rectangles). The definition uses closed cubes rather than closed rectangles purely for concretion. Using rectangles in place of cubes gives an identical set function (the two notions of covering are interchangeable, since every rectangle can be covered by finitely many cubes, and vice versa). Open cubes could equally be used in place of closed cubes, again without changing the value of m^* .

Example 5.1

$m^*({x}) = 0$ for any single point $x \in \mathbb{R}^d$: cover $\{x\}$ by a closed cube of side length ε centered at x , of volume ε^d . Since $\varepsilon > 0$ is arbitrary, the infimum over such covers is 0.

Remark 5.4. Since $m^*(E) = \inf_{E \subseteq \bigcup_{j=1}^{\infty} Q_j} \sum_{j=1}^{\infty} |Q_j|$, by the definition of infimum, for every $\varepsilon > 0$, we can find a cover using closed cubes $E \subseteq \bigcup_{j=1}^{\infty} Q_j$ such that

$$\sum_{j=1}^{\infty} |Q_j| \leq m^*(E) + \varepsilon. \quad (5.2)$$

This will turn out to be helpful while proving some identities.

We record some basic properties of m^* . These would guide us toward the properties we ultimately want a measure to have.

Proposition 5.1 (Monotonicity)

If $E \subseteq F \subseteq \mathbb{R}^d$, then $m^*(E) \leq m^*(F)$.

Proof. Every countable cover of F by closed cubes is also a cover of E (since $E \subseteq F$), so the collection of covers admissible for computing $m^*(F)$ is a subset of those admissible for $m^*(E)$. An infimum taken over a smaller collection of covers cannot be less than one taken over a larger collection, i.e. $m^*(E) \leq m^*(F)$. ■

Proposition 5.2 (Countable subadditivity)

For any countable collection of sets $E_1, E_2, \dots \subseteq \mathbb{R}^d$,

$$m^*\left(\bigcup_{j=1}^{\infty} E_j\right) \leq \sum_{j=1}^{\infty} m^*(E_j). \quad (5.3)$$

Proof. If the right side is infinite there is nothing to prove, so assume it is finite. Fix $\varepsilon > 0$. For each j , by definition of $m^*(E_j)$ as an infimum, there is a countable cover of E_j by closed cubes $\{Q_{j,i}\}_{i \geq 1}$ with

$$\sum_{i=1}^{\infty} |Q_{j,i}| \leq m^*(E_j) + \frac{\varepsilon}{2^j}. \quad (5.4)$$

The combined collection $\{Q_{j,i}\}_{j,i \geq 1}$ is a countable cover of $\bigcup_j E_j$ by closed cubes, so by definition of $m^*\left(\bigcup_j E_j\right)$ as an infimum,

$$m^*\left(\bigcup_{j=1}^{\infty} E_j\right) \leq \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} |Q_{j,i}| \leq \sum_{j=1}^{\infty} \left(m^*(E_j) + \frac{\varepsilon}{2^j}\right) = \sum_{j=1}^{\infty} m^*(E_j) + \varepsilon. \quad (5.5)$$

Since $\varepsilon > 0$ was arbitrary, the claimed inequality follows. ■

Corollary 5.3

If E is countable, then $m^*(E) = 0$.

Proof. Suppose $E = \{x_1, x_2, x_3, \dots\}$. Then

$$E = \bigcup_{i=1}^{\infty} \{x_i\} \implies 0 \leq m^*(E) \leq \sum_{i=1}^{\infty} m^*(\{x_i\}) = 0. \quad (5.6)$$

■

Proposition 5.4 (Exterior measure of a cube equals its volume)

If Q is a closed cube, then $m^*(Q) = |Q|$.

Proof. The single-cube cover $\{Q\}$ is admissible, giving $m^*(Q) \leq |Q|$ directly from the definition.

Now we shall prove that $m^*(Q) \geq |Q|$. Let $\{Q_j\}_{j \geq 1}$ be any countable cover of Q by closed cubes; we must show $\sum_{j=1}^{\infty} |Q_j| \geq |Q|$.

Fix $\varepsilon > 0$, and for each j let $\tilde{Q}_j^\circ$ be the open cube with the same center as Q_j , dilated by a factor $(1 + \varepsilon)$, i.e.

$$\tilde{Q}_j^\circ = (1 + \varepsilon) Q_j^\circ, \quad (5.7)$$

so that $Q_j \subset \tilde{Q}_j^\circ$ and

$$|\tilde{Q}_j^\circ| = |\tilde{Q}_j| = (1 + \varepsilon)^d |Q_j|. \quad (5.8)$$

Then $\{\tilde{Q}_j^\circ\}$ is a cover of Q by open sets, and since Q is compact, finitely many of them already cover Q , say $Q \subseteq \bigcup_{j=1}^n \tilde{Q}_j^\circ$. By [Proposition 4.2](#) (subadditivity of volume for coverings of a rectangle),

$$|Q| \leq \sum_{j=1}^n |\tilde{Q}_j^\circ| = (1 + \varepsilon)^d \sum_{j=1}^n |Q_j| \leq (1 + \varepsilon)^d \sum_{j=1}^{\infty} |Q_j|. \quad (5.9)$$

Since $\varepsilon > 0$ is arbitrary, letting $\varepsilon \rightarrow 0^+$ gives $|Q| \leq \sum_{j=1}^{\infty} |Q_j|$. As this holds for every admissible cover $\{Q_j\}$, taking the infimum over all such covers gives $|Q| \leq m^*(Q)$. ■

Proposition 5.5

If Q° is an open cube, then $m^*(Q^\circ) = |Q^\circ|$.

Proof. Let Q be the closure of Q° . Then $Q^\circ \subseteq Q$ and monotonicity ([Proposition 5.1](#)) gives us

$$m^*(Q^\circ) \leq m^*(Q) = |Q| = |Q^\circ|. \quad (5.10)$$

Now note that $(1 - \varepsilon)Q \subseteq Q^\circ$, and hence, monotonicity will give us

$$m^*(Q^\circ) \geq m^*((1 - \varepsilon)Q) = |(1 - \varepsilon)Q| = (1 - \varepsilon)^d |Q| = (1 - \varepsilon)^d |Q^\circ|. \quad (5.11)$$

Letting $\varepsilon \rightarrow 0^+$ forces $m^*(Q^\circ) \geq |Q^\circ|$. ■

The same argument (covering by cubes and using the volume-subadditivity [Proposition 4.2](#)) shows more generally that $m^*(R) = |R|$ for any rectangle R .

Proposition 5.6

For any rectangle R , $m^*(R) = |R|$.

Proof. Let $\{Q_j\}_{j \geq 1}$ be any countable cover of R by closed cubes. Fix $\varepsilon > 0$, and for each j let

$$\tilde{Q}_j^\circ = (1 + \varepsilon) Q_j^\circ, \quad (5.12)$$

so that $R \subseteq \bigcup_{j=1}^\infty Q_j \subseteq \bigcup_{j=1}^\infty \tilde{Q}_j^\circ$. By the compactness of R , we have $R \subseteq \bigcup_{j=1}^n \tilde{Q}_j^\circ$. Using [Proposition 4.2](#), we get

$$|R| \leq \sum_{j=1}^n |\tilde{Q}_j^\circ| = (1 + \varepsilon)^d \sum_{j=1}^n |Q_j| \leq (1 + \varepsilon)^d \sum_{j=1}^\infty |Q_j|. \quad (5.13)$$

Since $\varepsilon > 0$ is arbitrary, letting $\varepsilon \rightarrow 0^+$ gives $|R| \leq \sum_{j=1}^\infty |Q_j|$. As this holds for every admissible cover $\{Q_j\}$, taking the infimum over all such covers gives $|R| \leq m^*(R)$.

Now we'll show that $m^*(R) \leq |R|$. Consider a grid in $\mathbb{R}^d$ formed by cubes of size $\frac{1}{k}$.

$$\mathcal{Q} := \{Q : Q \text{ is a cube contained in } R\}, \quad (5.14)$$

$$\mathcal{Q}' := \{Q : Q \text{ is a cube that intersect both } R \text{ and } R^c\}. \quad (5.15)$$

Then we have

$$\bigcup_{Q \in \mathcal{Q}} Q \subseteq R \subseteq \bigcup_{Q \in \mathcal{Q} \cup \mathcal{Q}'} Q. \quad (5.16)$$

R is made out of $\bigcup_{Q \in \mathcal{Q}} Q$ (which itself is a collection of almost-disjoint rectangles) and some more almost-disjoint rectangles. Therefore,

$$|R| = \sum_{Q \in \mathcal{Q}} |Q| + \text{possibly some more} \geq \sum_{Q \in \mathcal{Q}} |Q|. \quad (5.17)$$

Claim 5.1: The size of $\mathcal{Q}'$ is at most Ck^{d-1} , for some constant C (not depending on k).

Proof. The cubes in $\mathcal{Q}'$ intersect both R and R^c , i.e. they cover the $(d-1)$ -dimensional faces of R . There are exactly $2d$ of them.

Let a be the largest edge-length of R . To cover the edge of length a , we would need around $\frac{a}{k-1} = ak$ cubes. But this would be the case if the cubes fit perfectly into the edge. There may be a little space leftover, so the number of cubes we need is no more than $ak + 2$ (adding one for each endpoint of the edge). Therefore, to cover a $(d-1)$ -dimensional face of R , the number of cubes needed is at most

$$(ak + 2)^{d-1} = (a + 1)^{d-1} k^{d-1}. \quad (5.18)$$

There are $2d$ faces, so the number of cubes needed to cover the faces is at most

$$2d \cdot (a + 1)^{d-1} k^{d-1}. \quad (5.19)$$

□

Then we have $R \subseteq \bigcup_{Q \in \mathcal{Q} \cup \mathcal{Q}'} Q$, and each cube has volume $\frac{1}{k^d}$. Therefore,

$$\sum_{Q \in \mathcal{Q} \cup \mathcal{Q}'} |Q| = \sum_{Q \in \mathcal{Q}} |Q| + \sum_{Q \in \mathcal{Q}'} |Q| \leq |R| + Ck^{d-1} \cdot \frac{1}{k^d} = |R| + \frac{C}{k}. \quad (5.20)$$

Taking $k \rightarrow \infty$, we get $m^*(R) \leq |R|$. ■

Proposition 5.7

For any $E \subseteq \mathbb{R}^d$,

$$m^*(E) = \inf\{m^*(O) : E \subseteq O, O \text{ open}\}. \quad (5.21)$$

Proof. For any open $O \supseteq E$, monotonicity (Proposition 5.1) gives $m^*(E) \leq m^*(O)$; taking the infimum over such O preserves the inequality:

$$m^*(E) \leq \inf_{\substack{E \subseteq O, \\ O \text{ open}}} m^*(O). \quad (5.22)$$

Now, for the other inequality: fix $\varepsilon > 0$. By definition of $m^*(E)$, there is a countable cover of E by closed cubes $\{Q_j\}$ with

$$\sum_{j=1}^{\infty} |Q_j| \leq m^*(E) + \varepsilon. \quad (5.23)$$

For each j , let

$$\tilde{Q}_j^\circ = (1 + \varepsilon) Q_j^\circ, \quad \text{and} \quad O := \bigcup_j \tilde{Q}_j^\circ, \quad (5.24)$$

which is an open set containing E (since $E \subseteq \bigcup_j Q_j \subseteq \bigcup_j \tilde{Q}_j^\circ$). By countable subadditivity (Proposition 5.2) and Proposition 5.4,

$$m^*(O) \leq \sum_{j=1}^{\infty} m^*(\tilde{Q}_j^\circ) = \sum_{j=1}^{\infty} |\tilde{Q}_j^\circ| \leq (1 + \varepsilon)^d \sum_{j=1}^{\infty} |Q_j| \leq (1 + \varepsilon)^d m^*(E) + \varepsilon. \quad (5.25)$$

Taking infimum on the LHS gives

$$\inf_{\substack{E \subseteq O, \\ O \text{ open}}} m^*(O) \leq (1 + \varepsilon)^d m^*(E) + \varepsilon, \quad (5.26)$$

for every $\varepsilon > 0$. Now taking $\varepsilon \rightarrow 0^+$ gives us the desired bound. ■

Definition 5.2. The diameter of a set $X \subseteq \mathbb{R}^d$ is

$$\text{diam}(X) = \sup\{|x - y| : x, y \in X\}. \quad (5.27)$$

The distance between two sets $X, Y \subseteq \mathbb{R}^d$ is

$$d(X, Y) := \inf\{|x - y| : x \in X, y \in Y\}. \quad (5.28)$$

Proposition 5.8 (Additivity for positively separated sets)

If $d(E_1, E_2) > 0$, then

$$m^*(E_1 \cup E_2) = m^*(E_1) + m^*(E_2). \quad (5.29)$$

Proof. Applying subadditivity (Proposition 5.2) to the two-set collection $\{E_1, E_2\}$, we get

$$m^*(E_1 \cup E_2) \leq m^*(E_1) + m^*(E_2). \quad (5.30)$$

For the other inequality, we may assume $m^*(E_1 \cup E_2) < \infty$ (else there is nothing to prove). Fix $\varepsilon > 0$ and let $\{Q_j\}$ be a cover of $E_1 \cup E_2$ by closed cubes with

$$\sum_{j=1}^{\infty} |Q_j| \leq m^*(E_1 \cup E_2) + \varepsilon. \quad (5.31)$$

By subdividing each Q_j into finitely many smaller closed cubes if necessary (which does not change the total volume, by Proposition 4.1), we may further assume every cube in the cover has diameter less than δ , with

$$0 < \delta < d(E_1, E_2). \quad (5.32)$$

Then no single cube Q_j can intersect both E_1 and E_2 ; indeed if Q_i intersects both E_1 and E_2 at points e_1 and e_2 , respectively, then

$$|e_1 - e_2| \leq \text{diam}(Q_j) \leq \delta < d(E_1, E_2) \leq |e_1 - e_2|, \quad (5.33)$$

which is not possible. Now, we partition the cover into $\{Q_j : Q_j \cap E_1 \neq \emptyset\}$, which covers E_1 , and $\{Q_j : Q_j \cap E_2 \neq \emptyset\}$, which covers E_2 (every cube meets at most one of the two sets, so this is a genuine partition of the index set). Then

$$m^*(E_1) \leq \sum_{Q_j \cap E_1 \neq \emptyset} |Q_j| \quad \text{and} \quad m^*(E_2) \leq \sum_{Q_j \cap E_2 \neq \emptyset} |Q_j|. \quad (5.34)$$

Adding these, we get

$$m^*(E_1) + m^*(E_2) \leq \sum_{Q_j \cap E_1 \neq \emptyset} |Q_j| + \sum_{Q_j \cap E_2 \neq \emptyset} |Q_j| = \sum_{j=1}^{\infty} |Q_j| \leq m^*(E_1 \cup E_2) + \varepsilon. \quad (5.35)$$

Since $\varepsilon > 0$ is arbitrary, $m^*(E_1) + m^*(E_2) \leq m^*(E_1 \cup E_2)$. ■

Corollary 5.9

If $E_1, E_2, \dots, E_n \subseteq \mathbb{R}^d$ with $d(E_i, E_j) > 0$ for every $i \neq j$, then

$$m^*\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n m^*(E_i). \quad (5.36)$$

Proposition 5.10 (Exterior measure of an almost disjoint union of cubes)

If $E = \bigcup_{j=1}^{\infty} Q_j$ is a countable, almost disjoint union of closed cubes, then

$$m^*(E) = \sum_{j=1}^{\infty} |Q_j|. \quad (5.37)$$

Proof. $m^*(E) \leq \sum_{j=1}^{\infty} |Q_j|$ directly follows from the definition, since $\bigcup_{j=1}^{\infty} Q_j$ is countable collection of closed cubes that cover E . For the other direction, fix $N \geq 1$ and $\varepsilon > 0$, and let

$$\tilde{Q}_j = (1 - \varepsilon) Q_j. \quad (5.38)$$

Since $Q_1, \dots, Q_N$ are almost disjoint (disjoint interiors) and there are only finitely many of them, shrinking creates a positive gap between each pair, i.e. for $i \neq j$,

$$d(Q'_i, Q'_j) > 0. \quad (5.39)$$

Corollary 5.9 together with Proposition 5.4 gives

$$m^*\left(\bigcup_{j=1}^N \tilde{Q}_j\right) = \sum_{j=1}^N m^*(\tilde{Q}_j) = \sum_{j=1}^N |\tilde{Q}_j| = (1 - \varepsilon)^d \sum_{j=1}^N |Q_j|. \quad (5.40)$$

Since $\bigcup_{j=1}^N Q'_j \subseteq E$, monotonicity (Proposition 5.1) gives

$$m^*(E) \geq m^*\left(\bigcup_{j=1}^N \tilde{Q}_j\right) = (1 - \varepsilon)^d \sum_{j=1}^N |Q_j|. \quad (5.41)$$

Letting $\varepsilon \rightarrow 0^+$ (with N fixed), so $m^*(E) \geq \sum_{j=1}^N |Q_j|$. Since this holds for every N , letting $N \rightarrow \infty$ gives $m^*(E) \geq \sum_{j=1}^{\infty} |Q_j|$. ■

Remark 5.5. Proposition 5.10 combined with Proposition 4.4 (every open set is a countable almost disjoint union of dyadic cubes), gives a concrete way of computing $m^*(O)$ for an open set O . That is, decompose O into almost disjoint dyadic cubes and sum their volumes.

We now single out the sets on which m^* will be shown (in the next lecture) to be countably additive.

Definition 5.3 (Lebesgue measurable set). A set $E \subseteq \mathbb{R}^d$ is **Lebesgue measurable** (or simply **measurable**, in this context) if, for every $\varepsilon > 0$, there exists an open set $O \supseteq E$ such that

$$m^*(O - E) \leq \varepsilon. \quad (5.42)$$

Remark 5.6. Informally, E is measurable if it can be approximated from outside by open sets, in the sense that the “error” $O - E$ can be made to have arbitrarily small exterior measure.

This definition makes no direct reference to a σ -algebra. We will show below that the collection of measurable sets does in fact form a σ -algebra.

Remark 5.7. [Proposition 5.7](#) says that for every $\varepsilon > 0$, we can find an open set $O \supseteq E$ such that

$$m^*(O) \leq m^*(E) + \varepsilon. \quad (5.43)$$

Note that this is quite different from [\(5.42\)](#), since it's **NOT** always true that $m^*(O - E) = m^*(O) - m^*(E)$.

We now establish a few basic properties of Lebesgue measurable sets, culminating in the fact that they form a σ -algebra.

Proposition 5.11 (Open sets are measurable)

Every open set $O \subseteq \mathbb{R}^d$ is measurable.

Proof. Take the open set to be O itself. Then $m^*(O - O) = m^*(\emptyset) = 0 < \varepsilon$ for every $\varepsilon > 0$. ■

Proposition 5.12 (Sets of exterior measure zero are measurable)

If $m^*(E) = 0$, then E is measurable. (In particular, $m^*(E) = 0$ forces E to be measurable with m^* restricting to a genuine value 0 on E ; such sets are called **null sets**.)

Proof. Fix $\varepsilon > 0$. By [Proposition 5.7](#) applied to E , since $m^*(E) = 0$, there is an open set $O \supseteq E$ with $m^*(O) \leq \varepsilon$. Since $O - E \subseteq O$, monotonicity gives

$$m^*(O - E) \leq m^*(O) \leq \varepsilon. \quad (5.44)$$

■

Proposition 5.13 (Countable unions of measurable sets are measurable)

If $E_1, E_2, \dots$ are measurable, then $E := \bigcup_{n=1}^{\infty} E_n$ is measurable.

Proof. Fix $\varepsilon > 0$. For each n , since E_n is measurable, choose an open set $O_n \supseteq E_n$ with

$$m^*(O_n - E_n) \leq \frac{\varepsilon}{2^n}. \quad (5.45)$$

Let $O := \bigcup_{n=1}^{\infty} O_n$, which is an open set containing E . Since

$$O - E \subseteq \bigcup_{n=1}^{\infty} (O_n - E_n) \quad (5.46)$$

(a point in O not in E lies in some O_n , and since it is not in $E \supseteq E_n$, it is not in E_n either, so it lies in $O_n - E_n$), countable subadditivity ([Proposition 5.2](#)) gives

$$m^*(O - E) \leq \sum_{n=1}^{\infty} m^*(O_n - E_n) \leq \sum_{n=1}^{\infty} \frac{\varepsilon}{2^n} = \varepsilon. \quad (5.47)$$

Proposition 5.14 (Closed sets are measurable)

Every closed set $F \subseteq \mathbb{R}^d$ is measurable.

Proof. Case 1: Compact Sets. Let K be compact (closed and bounded), we'll show that it's measurable. Fix $\varepsilon > 0$. By [Proposition 5.7](#), there is an open set $O \supseteq K$ with

$$m^*(O) \leq m^*(K) + \varepsilon. \quad (5.48)$$

Since K is bounded, $m^*(K) < \infty$, so this is a genuine (finite) bound. The set $O - K$ is open (as K is closed), so by [Proposition 4.4](#) it may be written as a countable almost disjoint union of closed cubes,

$$O - K = \bigcup_{n=1}^{\infty} Q_n. \quad (5.49)$$

For each fixed N , the finite union $S_N := \bigcup_{n=1}^N Q_n$ is compact (a finite union of closed bounded cubes), disjoint from K (since each $Q_n \subseteq O - K$), and two disjoint compact sets have positive distance; hence $d(K, S_N) > 0$ (see the following claim).

Claim 5.2: If $X, Y \subseteq \mathbb{R}^d$ are disjoint sets with X being compact and Y closed, then $d(X, Y) > 0$.

Proof. Assume for the sake of contradiction that there exists two sequences $x_n \in X$ and $y_n \in Y$ such that $|x_n - y_n| \rightarrow 0$. X is compact, so there exists a convergent subsequence $x_{n_i} \rightarrow x \in X$. Then

$$|x - y_{n_i}| \leq |x - x_{n_i}| + |x_{n_i} - y_{n_i}|. \quad (5.50)$$

Both terms of RHS converges to 0. Therefore $|x - y_{n_i}| \rightarrow 0$. As Y is closed, it contains all the limit points, so $x \in Y$. But we are given that X and Y are disjoint. Contradiction! $\square$

By [Proposition 5.8](#) and [Proposition 5.10](#),

$$m^*\left(K \cup \bigcup_{n=1}^N Q_n\right) = m^*(K) + m^*\left(\bigcup_{n=1}^N Q_n\right) = m^*(K) + \sum_{n=1}^N |Q_n|. \quad (5.51)$$

Since $K \cup \bigcup_{n=1}^N Q_n \subseteq O$, monotonicity gives

$$m^*(K) + \sum_{n=1}^N |Q_n| \leq m^*(O) \leq m^*(K) + \varepsilon \implies \sum_{n=1}^N |Q_n| \leq \varepsilon, \quad (5.52)$$

for every N , and letting $N \rightarrow \infty$,

$$\sum_{n=1}^{\infty} |Q_n| \leq \varepsilon. \quad (5.53)$$

By [Proposition 5.10](#) again,

$$m^*(O - K) = m^*\left(\bigcup_{n=1}^{\infty} Q_n\right) = \sum_{n=1}^{\infty} |Q_n| \leq \varepsilon. \quad (5.54)$$

Since $\varepsilon > 0$ was arbitrary, K is measurable.

Case 2: General closed sets. Let F be closed (not necessarily bounded). Write

$$F = \bigcup_{n=1}^{\infty} \left(F \cap \overline{B(\mathbf{0}, n)} \right), \quad (5.55)$$

a countable union of compact sets (each $F \cap \overline{B(\mathbf{0}, n)}$ is closed and bounded). By Case 1, each is measurable, and by [Proposition 5.13](#), their countable union F is measurable. ■

Proposition 5.15 (Complements of measurable sets are measurable)

If $E \subseteq \mathbb{R}^d$ is measurable, then $E^c = \mathbb{R}^d - E$ is measurable.

Proof. For each $n \geq 1$, since E is measurable, choose an open set $O_n \supseteq E$ with

$$m^*(O_n - E) \leq \frac{1}{n}. \quad (5.56)$$

Each O_n^c is closed, hence measurable by [Proposition 5.14](#). Moreover, $O_n^c \subseteq E^c$, since $E \subseteq O_n$. Let

$$S := \bigcup_{n=1}^{\infty} O_n^c, \quad (5.57)$$

a countable union of measurable sets, hence measurable by [Proposition 5.13](#). Since each $O_n^c \subseteq E^c$, also $S \subseteq E^c$. We claim $m^*(E^c - S) = 0$. Indeed,

$$E^c - S = E^c \cap S^c = E^c \cap \bigcap_{n=1}^{\infty} O_n = \bigcap_{n=1}^{\infty} (E^c \cap O_n) \subseteq E^c \cap O_n = O_n - E, \quad (5.58)$$

for every n . so

$$m^*(E^c - S) \leq m^*(O_n - E) \leq \frac{1}{n}, \quad (5.59)$$

for every n , forcing $m^*(E^c - S) = 0$. By [Proposition 5.12](#), $E^c - S$ is measurable. Finally,

$$E^c = S \cup (E^c - S) \quad (5.60)$$

is a union of two measurable sets, hence measurable by [Proposition 5.13](#). ■

Theorem 5.16 (The Lebesgue σ -algebra)

The collection $\mathcal{L}$ of Lebesgue measurable subsets of $\mathbb{R}^d$ is a σ -algebra.

Proof. [Proposition 5.13](#) gives closure of $\mathcal{L}$ under countable unions, and [Proposition 5.15](#) gives closure under complements. Furthermore, $\emptyset \in \mathcal{L}$ as it's a null-set. These are exactly the defining properties of a σ -algebra ([Definition 1.1](#)). ■

Remark 5.8. $\mathcal{L}$ contains every open set and hence contains the Borel σ -algebra $\mathcal{B}(\mathbb{R}^d)$. So in general

$$\mathcal{B}(\mathbb{R}^d) \subseteq \mathcal{L} \subsetneq \mathcal{P}(\mathbb{R}^d), \quad (5.61)$$

with the last containment strict by Vitali's theorem ([Theorem 3.11](#)).

Definition 5.4 (Lebesgue measure). The **Lebesgue measure** of a set $E \in \mathcal{L}$ is defined as

$$m(E) := m^*(E), \quad (5.62)$$

i.e. m is the restriction of the exterior measure m^* to the σ -algebra $\mathcal{L}$ of measurable sets.

It remains to verify that m is in fact countably additive on $\mathcal{L}$, i.e. that $(\mathbb{R}^d, \mathcal{L}, m)$ is genuinely a measure space in the sense of [Definition 1.3](#).

§6 18 September 2026

The first order of business is to show that the set function m just defined is in fact a genuine *measure*: that it is countably additive on disjoint measurable sets.

Lemma 6.1

If E is measurable, then for every $\varepsilon > 0$, there exists a closed set $F \subseteq E$ such that $m^*(E - F) \leq \varepsilon$.

Proof. E^c is measurable, so there is an open set $O \supseteq E^c$ with $m(O - E^c) \leq \varepsilon$. Let $F = O^c$; then F is closed, and $F = O^c \subseteq (E^c)^c = E$, while

$$E - F = E \cap O = O - E^c, \quad (6.1)$$

so $m(E - F) = m(O - E^c) \leq \varepsilon$. ■

Theorem 6.2 (Countable additivity)

If $E_1, E_2, E_3, \dots$ is a countable collection of pairwise disjoint measurable sets, and $E = \bigcup_{j=1}^{\infty} E_j$, then E is measurable and

$$m(E) = \sum_{j=1}^{\infty} m(E_j). \quad (6.2)$$

Proof. Measurability of $E = \bigcup_j E_j$ is immediate, since $\mathcal{L}$ is already known to be a σ -algebra (closed under countable unions). Also, since the exterior measure is already known to be countably *sub*additive,

$$m(E) \leq \sum_{j=1}^{\infty} m(E_j). \quad (6.3)$$

So we'll now prove the other inequality. First, assume each E_j is bounded. Fix $\varepsilon > 0$. Since each E_j is measurable, there is a closed set $F_j \subseteq E_j$ with

$$m^*(E_j - F_j) \leq \frac{\varepsilon}{2^j}. \quad (6.4)$$

Since $E_j - F_j$ and F_j are disjoint with union E_j , using subadditivity,

$$m^*(E_j) \leq m^*(F_j) + m^*(E_j - F_j) \leq m^*(F_j) + \frac{\varepsilon}{2^j} \implies m(F_j) \geq m(E_j) - \frac{\varepsilon}{2^j}. \quad (6.5)$$

Now fix N and consider $F_1, \dots, F_N$. Since $E_1, \dots, E_N$ are pairwise disjoint, so are $F_1, \dots, F_N$. Moreover, each F_j is closed and bounded (as $F_j \subseteq E_j$ and E_j is bounded), hence compact.

Applying [Corollary 5.9](#) and [Claim 5.2](#),

$$m^*\left(\bigcup_{j=1}^N F_j\right) = \sum_{j=1}^N m(F_j). \quad (6.6)$$

Since $\bigcup_{j=1}^N F_j \subseteq E$, monotonicity of outer measure gives

$$m(E) \geq m^*\left(\bigcup_{j=1}^N F_j\right) = \sum_{j=1}^N m(F_j) \geq \sum_{j=1}^N \left(m(E_j) - \frac{\varepsilon}{2^j}\right) \geq \sum_{j=1}^N m(E_j) - \varepsilon. \quad (6.7)$$

Letting $N \rightarrow \infty$ and then $\varepsilon \rightarrow 0$ (which was arbitrary) yields

$$m(E) \geq \sum_{j=1}^{\infty} m(E_j), \quad (6.8)$$

as desired. Now we move to the general scenario of when E_j 's need not be bounded. Write $\mathbb{R}^d$ as an increasing union of closed balls: let B_k be the closed ball of radius k centered at the origin, and set

$$S_1 = B_1, \quad S_k = B_k - B_{k-1} \quad (k \geq 2), \quad (6.9)$$

so that the S_k are pairwise disjoint and $\mathbb{R}^d = \bigcup_{k=1}^{\infty} S_k$. For each j , decompose E_j according to this partition:

$$E_{j,k} = E_j \cap S_k, \quad E_j = \bigcup_{k=1}^{\infty} E_{j,k}, \quad (6.10)$$

a disjoint union of *bounded* measurable sets. By applying the bounded case to the sets $\{E_{j,k}\}_k$ (disjoint and bounded, for fixed j),

$$m(E_j) = \sum_{k=1}^{\infty} m(E_{j,k}). \quad (6.11)$$

Now $E = \bigcup_j E_j = \bigcup_{j,k} E_{j,k}$, and all the sets $E_{j,k}$ (over all j and k) are pairwise disjoint and bounded (each being contained in S_k), so the bounded case applies again to this doubly-indexed countable disjoint bounded family:

$$m(E) = \sum_{j,k} m(E_{j,k}) = \sum_{j=1}^{\infty} \left(\sum_{k=1}^{\infty} m(E_{j,k}) \right) = \sum_{j=1}^{\infty} m(E_j). \quad (6.12)$$

■

Remark 6.1. The key point used in this argument was that each E_j could be approximated from inside by a *compact* set F_j . Disjointness of the E_j then translates into a positive distance between the compact pieces F_j , which is what makes the finite additivity of outer measure on compact sets available.

Corollary 6.3

If $F \subseteq E$ are both measurable with $m(F) < \infty$, then

$$m(E - F) = m(E) - m(F). \quad (6.13)$$

Proof. $E = F \cup (E - F)$, and they are disjoint. So by additivity,

$$m(E) = m(F) + m(E - F). \quad (6.14)$$

Since $m(F) < \infty$, we can subtract it from both sides and the desired equality follows. $\blacksquare$

Proposition 6.4

Let $E \subseteq \mathbb{R}^d$ be Lebesgue measurable. Then the following hold.

- (i) If $m(E) < \infty$, then for every $\varepsilon > 0$ there exists a compact set $K \subseteq E$ with

$$m(E - K) \leq \varepsilon. \quad (6.15)$$

- (i) If $m(E) < \infty$, then for every $\varepsilon > 0$ there exists a finite union $F = \bigcup_{i=1}^N Q_i$ of almost disjoint closed cubes such that

$$m(E \triangle F) \leq \varepsilon, \quad (6.16)$$

where $E \triangle F = (E - F) \cup (F - E)$ denotes the symmetric difference.

Proof. (i) Write E as the disjoint union of its intersections with the shells S_k from earlier:

$$E = \bigcup_{k=1}^{\infty} E_k, \quad E_k = E \cap S_k. \quad (6.17)$$

By countable additivity, $m(E) = \sum_{k=1}^{\infty} m(E_k)$. Since $m(E) < \infty$ this series converges, so there is N such that

$$\sum_{k=1}^N m(E_k) \geq m(E) - \frac{\varepsilon}{2}. \quad (6.18)$$

Since $\bigcup_{k=1}^N S_k = B_n$ (the closed ball of radius N),

$$m(E \cap B_n) = \sum_{k=1}^n m(E_k) \geq m(E) - \frac{\varepsilon}{2}. \quad (6.19)$$

Now, by applying [Lemma 6.1](#) to $E \cap B_n$, there is a closed set $K \subseteq E \cap B_n$ with

$$m((E \cap B_n) - K) \leq \frac{\varepsilon}{2}. \quad (6.20)$$

Since $K \subseteq B_n$, K is closed and bounded, hence compact, and $K \subseteq E$. Finally

$$E - K = (E - (E \cap B_n)) \cup ((E \cap B_n) - K), \quad (6.21)$$

a disjoint union, so

$$m(E - K) = m(E) - m(E \cap B_n) + m((E \cap B_n) - K) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad (6.22)$$

- (ii) Fix an open set $O \supseteq E$ with $m(O - E) < \varepsilon/2$. In particular, $m(O) < \infty$. Every open subset of $\mathbb{R}^d$ can be written as a countable union of almost disjoint closed cubes,

$$O = \bigcup_{j=1}^{\infty} Q_j, \quad \{Q_j\} \text{ almost disjoint}, \quad (6.23)$$

and by countable additivity of almost-disjoint cubes ([Proposition 5.10](#)),

$$m(O) = \sum_{j=1}^{\infty} |Q_j| < \infty. \quad (6.24)$$

Choose N so that

$$\sum_{j=1}^N |Q_j| \geq m(O) - \frac{\varepsilon}{2}, \quad (6.25)$$

and set $F = \bigcup_{j=1}^N Q_j$, a finite union of almost disjoint closed cubes with $F \subseteq O$ and

$$m(O - F) = m(O) - m(F) \leq \frac{\varepsilon}{2}. \quad (6.26)$$

Now estimate the symmetric difference $E \triangle F = (E - F) \cup (F - E)$: Since $E \subseteq O$, we have $E - F \subseteq O - F$, so

$$m(E - F) \leq m(O - F) \leq \frac{\varepsilon}{2}. \quad (6.27)$$

Since $F \subseteq O$, we have $F - E \subseteq O - E$, so

$$m(F - E) \leq m(O - E) \leq \frac{\varepsilon}{2}. \quad (6.28)$$

Adding, $m(E \triangle F) \leq m(E - F) + m(F - E) \leq \varepsilon$. ■

Remark 6.2. These statements express the same intuitive fact in different guises. A measurable set can be approximated from outside by an open set, from inside by a closed set, from inside by a compact set (when it has finite measure), and by a finite union of almost-disjoint cubes, up to a set of arbitrarily small measure. One chooses whichever form of approximation is most convenient for the problem at hand.

One can show that $(\mathbb{R}^d, \mathcal{L}, m)$ is a complete measure space.

Theorem 6.5

m is a complete measure.

Proof. Suppose $E \in \mathcal{L}$ with $m(E) = 0$. Then for any subset $A \subseteq E$, by the monotonicity of exterior measure, $m^*(A) \leq m^*(E) = 0$, implying that A is also a null-set. Since null-sets are measurable, $A \in \mathcal{L}$. So m is a complete measure. ■

The Lebesgue measurable sets arise as the completion of the σ -algebra of Borel sets, that is by adjoining all subsets of Borel sets of measure zero.

Definition 6.1. A set is called G_δ set if it is the intersection of a countable collection of open sets. A set is called F_σ set if it is the union of a countable collection of closed sets.

Proposition 6.6

Let $E \subseteq \mathbb{R}^d$. The following are equivalent:

- (a) E is measurable.
- (b) There is a G_δ -set $G \supseteq E$ such that $m^*(G - E) = 0$.
- (c) There is a F_σ -set $F \subseteq E$ such that $m^*(E - F) = 0$.

Proof. **(b) $\Rightarrow$ (a)** G is measurable, since it's the intersection of countably many measurable sets. So is $G - E$, since it's a null-set. Therefore, $E = G - (G - E)$ is a measurable set.

(c) $\Rightarrow$ (a) F is measurable, since it's the union of countably many measurable sets. So is $E - F$, since it's a null-set. Therefore, $E = F \cup (E - F)$ is a measurable set.

(a) $\Rightarrow$ (b) Since E is measurable, for every n , there exists an open set $O_n \supseteq E$ such that

$$m^*(O_n - E) \leq \frac{1}{n}. \quad (6.29)$$

Choose $G = \bigcap_{n=1}^{\infty} O_n \supseteq E$. Then,

$$G - E = \bigcap_{n=1}^{\infty} O_n - E \subseteq O_n - E \quad (6.30)$$

$$\implies m^*(G - E) \leq m^*(O_n - E) \leq \frac{1}{n}, \quad (6.31)$$

for every n . Therefore, $m^*(G - E) = 0$.

(a) $\Rightarrow$ (c) Since E is measurable, for every n , there exists an open set $F_n \subseteq E$ such that

$$m^*(E - F_n) \leq \frac{1}{n}. \quad (6.32)$$

Choose $F = \bigcup_{n=1}^{\infty} F_n \subseteq E$. Then,

$$E - F = E - \bigcap_{n=1}^{\infty} F_n \subseteq E - F_n \quad (6.33)$$

$$\implies m^*(E - F) \leq m^*(E - F_n) \leq \frac{1}{n}, \quad (6.34)$$

for every n . Therefore, $m^*(E - F) = 0$. ■

Proposition 6.7

$E \subseteq \mathbb{R}^d$ is measurable if and only if for every $\varepsilon > 0$, there exists $F \subseteq E \subseteq U$ with U open and F closed such that $m^*(U - F) \leq \varepsilon$.

Proof. Suppose E is measurable. Then there exists $F \subseteq E \subseteq U$ with U open and F closed such that

$$m^*(U - E) \leq \frac{\varepsilon}{2} \text{ and } m^*(E - F) \leq \frac{\varepsilon}{2}. \quad (6.35)$$

Then $U - F = (U - E) \cup (E - F)$ is a disjoint union, and hence

$$m^*(U - F) \leq m^*(U - E) + m^*(E - F) \leq \varepsilon. \quad (6.36)$$

Conversely, suppose such $F \subseteq E \subseteq U$ exists for every ε . In particular, for every n , we can choose $F_n \subseteq E \subseteq U_n$ with F_n closed and U_n open, and

$$m^*(U_n - F_n) \leq \frac{1}{n}. \quad (6.37)$$

Choose $F = \bigcup_{n=1}^{\infty} F_n$ is a F_σ set. Then we have $F \subseteq E$. Furthermore,

$$E - F = E - \bigcup_{n=1}^{\infty} F_n \subseteq E - F_n \subseteq U_n - F_n. \quad (6.38)$$

By monotonicity, we have

$$m^*(E - F) \leq m^*(U_n - F_n) \leq \frac{1}{n}, \quad (6.39)$$

which is true for every n . Therefore, $m^*(E - F) = 0$. This gives E is measurable by [Proposition 6.6](#). ■

Now we'll discuss some invariance properties of Lebesgue measure.

Theorem 6.8 (Translation invariance)

Let $E \subseteq \mathbb{R}^d$ be measurable and $x_0 \in \mathbb{R}^d$. Then $E + x_0 := \{x + x_0 : x \in E\}$ is measurable and

$$m(E + x_0) = m(E). \quad (6.40)$$

Proof. We divide the proof into several steps.

Claim 6.1: Outer measure is translation invariant.

Proof. Every covering of E by cubes $\{Q_j\}$ gives, upon translating each cube by x_0 , a covering of $E + x_0$ by cubes $\{Q_j + x_0\}$ of the *same* volumes (translating a cube does not change its side lengths). Hence

$$m^*(E + x_0) = m^*(E) \quad (6.41)$$

for every set E , since the coverings of E and of $E + x_0$ by cubes are in volume-preserving bijection with each other. $\square$

Claim 6.2: If E is measurable, so is $E + x_0$.

Proof. If E is measurable, for $\varepsilon > 0$ choose an open $O \supseteq E$ with $m^*(O - E) \leq \varepsilon$. Then $O + x_0$ is open, contains $E + x_0$, and

$$(O + x_0) - (E + x_0) = (O - E) + x_0, \quad (6.42)$$

so by [Claim 6.1](#),

$$m^*((O + x_0) - (E + x_0)) = m^*(O - E) \leq \varepsilon. \quad (6.43)$$

$\square$

Hence $E + x_0$ is measurable, and its measure equals $m^*(E + x_0) = m^*(E) = m(E)$. $\blacksquare$

Theorem 6.9 (Scaling/dilation invariance)

Let $E \subseteq \mathbb{R}^d$ be measurable and $\lambda > 0$. Then $\lambda E := \{\lambda x : x \in E\}$ is measurable and

$$m(\lambda E) = \lambda^d m(E). \quad (6.44)$$

Proof. As in the translation case, we divide the proof into several steps.

Claim 6.3: $m^*(\lambda E) = \lambda^d m^*(E)$ for every $E \subseteq \mathbb{R}^d$.

Proof. If $\{Q_j\}$ is a covering of E by cubes, then $\{\lambda Q_j\}$ covers λE , and a cube of side length ℓ scales to a cube of side length $\lambda\ell$, so its volume scales by λ^d . Hence

$$m^*(\lambda E) = \lambda^d m^*(E), \quad (6.45)$$

for every set E , since the coverings of E and of λE by cubes are in volume-scaling-by- λ^d bijection with each other. $\square$

Claim 6.4: If E is measurable, so is λE .

Proof. If E is measurable, choosing an open $O \supseteq E$ with $m^*(O - E) < \frac{\varepsilon}{\lambda^d}$. The set λO is open, contains λE , and $(\lambda O) - (\lambda E) = \lambda(O - E)$, so

$$m^*(\lambda O - \lambda E) = \lambda^d m^*(O - E) \leq \varepsilon, \quad (6.46)$$

as desired. $\square$

Hence, λE is measurable with $m(\lambda E) = \lambda^d m(E)$. $\blacksquare$

One can similarly prove a rotation invariance. However one can prove a more general fact.

Theorem 6.10

Let $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a linear map. If $E \subseteq \mathbb{R}^d$ is measurable, so is $T(E)$. Furthermore,

$$m(T(E)) = |\det T| m(E). \quad (6.47)$$

We shall prove it in several steps.

Proposition 6.11

Let $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a linear map with $\det T = 0$. Then $m^*(T(\mathbb{R}^d)) = 0$.

Proof. Let $H = T(\mathbb{R}^d)$. We can write

$$H = \bigcup_{R=1}^{\infty} (H \cap [-R, R]^d) =: \bigcup_{R=1}^{\infty} H_R. \quad (6.48)$$

By countable additivity,

$$m^*(H) \leq \sum_{R=1}^{\infty} m^*(H \cap [-R, R]^d). \quad (6.49)$$

So we will prove that $m^*(H \cap [-R, R]^d) = 0$ for every R . Fix N , and let's divide each side of $[-R, R]^d$ into N equal parts, i.e. divide the cube into N^d small cubes, each of side length $\frac{2R}{N}$. Let

$$\mathcal{Q} = \left\{ Q : Q \text{ is a small cube that intersects } H \cap [-R, R]^d \right\}. \quad (6.50)$$

Then $\mathcal{Q}$ covers $H \cap [-R, R]^d$, so

$$m^*(H \cap [-R, R]^d) \leq \sum_{Q \in \mathcal{Q}} |Q| = |\mathcal{Q}| \left(\frac{2R}{N} \right)^d, \quad (6.51)$$

where $|\mathcal{Q}|$ denotes the size of the set $\mathcal{Q}$. ¹

¹**Abuse of notation:** For a cube Q , $|Q|$ is its volume, and for the set $\mathcal{Q}$, $|\mathcal{Q}|$ is its cardinality.

Claim 6.5: $|\mathcal{Q}| \leq C \cdot N^{d-1}$ for some constant C (not depending on N).

Proof. Since H is a strictly lower-dimensional hyperplane through the origin, there exists some nonzero vector $\mathbf{v} \in \mathbb{R}^d$ such that $\mathbf{x} \cdot \mathbf{v} = 0$ for every $\mathbf{x} \in H$. WLOG we can assume $v_d \neq 0$ (well, there is a coordinate $v_{i_0} \neq 0$, then d can be replaced with i_0 the following argument). Consider a “column”

$$C = I_1 \times I_2 \times \cdots \times I_{d-1} \times [-R, R], \quad (6.52)$$

where I_i are intervals of length $\frac{2R}{N}$ in each direction. There are N^{d-1} such columns. Therefore,

$$|\mathcal{Q}| \leq N^{d-1} \times (\text{maximum number of cubes in a column that intersect } H_R). \quad (6.53)$$

If $\mathbf{y}, \mathbf{z} \in H_R$ that belong to the same column, then

$$\mathbf{y} \cdot \mathbf{v} = 0 \implies y_d = - \sum_{i=1}^{d-1} \frac{v_i}{v_d} y_i, \quad (6.54)$$

$$\mathbf{z} \cdot \mathbf{v} = 0 \implies z_d = - \sum_{i=1}^{d-1} \frac{v_i}{v_d} z_i, \quad (6.55)$$

$$\implies |y_d - z_d| \leq \left| \sum_{i=1}^{d-1} \frac{v_i}{v_d} (y_i - z_i) \right| \leq \sum_{i=1}^{d-1} \left| \frac{v_i}{v_d} \right| \frac{2R}{N}, \quad (6.56)$$

since $\mathbf{y}$ and $\mathbf{z}$'s i -th coordinate differs by at most $\frac{2R}{N}$. Hence, if $\mathbf{y}, \mathbf{z} \in H_R$ that belong to the same column, then their d -th coordinate differs by at most $M \cdot \frac{2R}{N}$, for some constant M that doesn't depend on N (but M may depend on T).

Therefore, we can argue that there are **at most** $M + 2$ cubes in a column that intersect H_R . Otherwise, we can find two cubes in a column, both of them intersect H_R , and there are at least $M + 1$ cubes in between them. This way, the distance between the d -th coordinate of those two points (at which the said cubes intersect H_R) would be at least $(M + 1) \frac{2R}{N}$.

Hence, $|\mathcal{Q}| \leq (M + 2) N^{d-1}$. □

Now, using (6.51),

$$m^* \left(H \cap [-R, R]^d \right) \leq |\mathcal{Q}| \left(\frac{2R}{N} \right)^d = C \cdot N^{d-1} \frac{(2R)^d}{N^d} = \frac{C'}{N}. \quad (6.57)$$

This is true for every N . By taking $N \rightarrow \infty$, we get $m^*(H_R) = 0$. By using subadditivity (6.49), we get that $m^*(H) = 0$. ■

This proves the $\det T = 0$ case of Theorem 6.10. Now, we focus on the $\det T \neq 0$ case. A standard linear algebra fact says that any non-singular linear map can be written as a composition of

“elementary” linear maps of the following types:

- (a) $T_1(x_1, \dots, x_i, \dots, x_d) = (x_1, \dots, cx_i, \dots, x_d)$ for $c \neq 0$;
- (b) $T_2(x_1, \dots, x_i, \dots, x_j, \dots, x_d) = (x_1, \dots, x_j, \dots, x_i, \dots, x_d)$.
- (c) $T_3(x_1, x_2, \dots, x_d) = (x_1 + x_2, x_2, \dots, x_d)$.

Also, note that

$$\det T_1 = c, \quad \det T_2 = -1, \quad \det T_3 = 1. \quad (6.58)$$

Lemma 6.12

Let $Q = [0, 1]^d$ be the unit cube. Then

- (a) $m^*(T_1(Q)) = |c|$;
- (b) $m^*(T_2(Q)) = 1$;
- (c) $m^*(T_3(Q)) \leq 1$.

In other words, $m^*(T_i(Q)) \leq |\det T_i|$ for each $i = 1, 2, 3$.

Proof. For T_1 , the image $T_1(Q)$ is the rectangle

$$[0, 1] \times \cdots \times [0, 1] \times [0, c] \times [0, 1] \times \cdots \times [0, 1], \quad (6.59)$$

or $[c, 0]$ instead of $[0, c]$. We know that the exterior measure of a rectangle is its volume (Proposition 5.6), hence,

$$m^*(T_1(Q)) = |c| = |\det T_1|. \quad (6.60)$$

For T_2 , the image $T_2(Q) = Q$, so

$$m^*(T_2(Q)) = m^*(Q) = 1 = |\det T_2|. \quad (6.61)$$

For T_3 , note that

$$T_3(Q) = \{(x_1, \dots, x_d) : 0 \leq x_i \leq 1 \text{ for } i \geq 2; \text{ and } x_2 \leq x_1 \leq x_2 + 1\}. \quad (6.62)$$

Note that $\tilde{Q} := T_3(Q)$ is contained in the rectangle $[0, 2] \times [0, 1]^{d-1}$. We cover this rectangle using cubes of side length $\frac{1}{N}$. Let

$$\mathcal{Q} = \{Q' : Q' \text{ is a small cube that intersects } \tilde{Q}\}. \quad (6.63)$$

Then $\mathcal{Q}$ covers $\tilde{Q}$, so

$$m^*(\tilde{Q}) \leq \sum_{Q' \in \mathcal{Q}} |Q'| = |\mathcal{Q}| \cdot \frac{1}{N^d}. \quad (6.64)$$

Again, $|\mathcal{Q}|$ denotes the size of the set $\mathcal{Q}$.

Claim 6.6: $|\mathcal{Q}| = (1 + N) N^{d-1}$.

Proof. Fix a “row”

$$R = [0, 2] \times I_2 \times I_3 \times \cdots \times I_d, \quad (6.65)$$

where each I_i is an interval of length $\frac{1}{N}$. There are N^{d-1} such rows. Now let's see how many small cubes of this row intersect $\tilde{Q}$.

Suppose $I_2 = [\frac{k}{N}, \frac{k+1}{N}]$ for some k . Now, take $\mathbf{x} \in \tilde{Q} \cap R$. Then

$$x_2 \leq x_1 \leq x_2 + 1 \implies x_1 \in \left[\frac{k}{N}, \frac{k + (N + 1)}{N} \right]. \quad (6.66)$$

Therefore, a small cube $Q' = I_1 \times I_2 \times \cdots \times I_d$ intersects $\tilde{Q}$ if and only if $I_1 \subseteq [\frac{k}{N}, \frac{k + (N + 1)}{N}]$. There are $N + 1$ choices for I_1 then. As a result, there are $(N + 1)$ cubes in R that intersect $\tilde{Q}$. Since this is the same for all rows, we have $|\mathcal{Q}| = (1 + N) N^{d-1}$. $\square$

Now, (6.64) gives us

$$m^*(T_3(Q)) \leq \frac{(1 + N) N^{d-1}}{N^d} = 1 + \frac{1}{N}. \quad (6.67)$$

Since this is true for every N , by taking $N \rightarrow \infty$, we get $m^*(T_3(Q)) \leq 1$. $\blacksquare$

So we have shown that under elementary linear maps T , we have

$$m^*(T(Q)) \leq |\det T| m^*(Q), \quad (6.68)$$

where $Q = [0, 1]^d$ is the unit cube.

Lemma 6.13

For **every** cube Q' and elementary linear map T ,

$$m^*(T(Q')) \leq |\det T| m^*(Q'). \quad (6.69)$$

Proof. If Q' is another cube, then it's just a dilation and translation of Q , i.e. $Q' = \delta Q + s$. Then

$$T(Q') = T(\delta Q + s) = \delta T(Q) + T(s). \quad (6.70)$$

By the dilation and translation invariance of outer measure,

$$m^*(T(Q')) = m^*(\delta T(Q) + T(s)) = \delta^d m^*(T(Q)) \quad (6.71)$$

$$\leq |\det T| \delta^d m^*(Q) = |\det T| m^*(\delta Q + s) \quad (6.72)$$

$$= |\det T| m^*(Q'). \quad (6.73)$$

$\blacksquare$

Lemma 6.14

For **every** open set $O \subseteq \mathbb{R}^d$ and elementary linear map T ,

$$m^*(T(O)) \leq |\det T| m^*(O). \quad (6.74)$$

Proof. Write O as a almost-disjoint countable union of cubes: $O = \bigcup_{i=1}^{\infty} Q_i$. Then

$$m^*(T(O)) = m^*\left(T\left(\bigcup_{i=1}^{\infty} Q_i\right)\right) = m^*\left(\bigcup_{i=1}^{\infty} T(Q_i)\right) \quad (6.75)$$

$$\leq \sum_{i=1}^{\infty} m^*(T(Q_i)) \quad (6.76)$$

$$\leq |\det T| \sum_{i=1}^{\infty} m^*(Q_i) = |\det T| m^*(O), \quad (6.77)$$

where the last equality follows from [Proposition 5.10](#). ■

Lemma 6.15

Let $G \subseteq \text{GL}(d, \mathbb{R})$ be the set

$$G = \left\{ T \in \text{GL}(d, \mathbb{R}) : m^*(T(O)) \leq |\det T| m^*(O) \text{ for every open } O \subseteq \mathbb{R}^d \right\}. \quad (6.78)$$

Then G is closed under composition.

Proof. A linear map is continuous, and hence, an invertible linear map is a homeomorphism $\mathbb{R}^d \rightarrow \mathbb{R}^d$. Therefore, if O is open and $S, T \in G$, then both $T(O)$ and $ST(O)$ are open.

Now, since $T(O)$ is open and $S \in G$, we have

$$m^*(ST(O)) \leq |\det S| m^*(T(O)). \quad (6.79)$$

Since $T \in G$ and O is open, we have

$$m^*(T(O)) \leq |\det T| m^*(O). \quad (6.80)$$

Combining these, we have

$$m^*(ST(O)) \leq |\det S| |\det T| m^*(O) = |\det(ST)| m^*(O), \quad (6.81)$$

implying that $ST \in G$. ■

Proposition 6.16

$G = \text{GL}(d, \mathbb{R})$.

Proof. Every invertible linear map $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ can be written as a composition of the elementary maps. By [Lemma 6.14](#), elementary maps are in G . Since G is closed under composition, we have $T \in G$ for every T . ■

In other words, for **every** $T \in \text{GL}(d, \mathbb{R})$, we have

$$m^*(T(O)) \leq |\det T| m^*(O). \quad (6.82)$$

Actually, we can prove the equality in [\(6.82\)](#).

Proposition 6.17

For every open set $O \subseteq \mathbb{R}^d$ and $T \in \text{GL}(d, \mathbb{R})$, we have

$$m^*(T(O)) = |\det T| m^*(O). \quad (6.83)$$

Proof. We already have $m^*(T(O)) \leq |\det T| m^*(O)$. Now note that

$$m^*(O) = m^*(T^{-1}(T(O))) \quad (6.84)$$

Since $T(O)$ is also open, and $T^{-1} \in \text{GL}(d, \mathbb{R})$, using [\(6.82\)](#),

$$m^*(O) = m^*(T^{-1}(T(O))) \leq |\det(T^{-1})| m^*(T(O)) = \frac{1}{|\det T|} m^*(T(O)). \quad (6.85)$$

Rearranging, we get

$$m^*(T(O)) \geq |\det T| m^*(O). \quad (6.86)$$

■

Proposition 6.18

For every $E \subseteq \mathbb{R}^d$,

$$m^*(T(E)) = |\det T| m^*(E). \quad (6.87)$$

Proof. We know from [Proposition 5.7](#) that

$$m^*(E) = \inf_{O \supseteq E, O \text{ open}} m^*(O). \quad (6.88)$$

Since T is a homeomorphism, $O \supseteq T(E)$ if and only if $O = T(U)$ for some open $U \supseteq E$.

$$m^*(T(E)) = \inf_{\substack{O \supseteq T(E), \\ O \text{ open}}} m^*(O) = \inf_{\substack{U \supseteq E, \\ U \text{ open}}} m^*(T(U)) \quad (6.89)$$

$$= |\det T| \inf_{\substack{U \supseteq E, \\ U \text{ open}}} m^*(U) = |\det T| m^*(E). \quad (6.90)$$

■

Now we are pretty much ready to finish the proof of [Theorem 6.10](#).

Proof of Theorem 6.10. Suppose E is measurable, and $\varepsilon > 0$. Then we can find an open set $O \supseteq E$ such that

$$m^*(O - E) \leq \frac{\varepsilon}{\det T}. \quad (6.91)$$

Since T is a homeomorphism, $T(O)$ is an open set containing $T(E)$, and $T(O) - T(E) = T(O - E)$. As a result,

$$m^*(T(O) - T(E)) = m^*(T(O - E)) = |\det T| m^*(O - E) \leq \varepsilon. \quad (6.92)$$

So $T(E)$ is measurable. Then $m(T(E)) = |\det T| m(E)$ follows from [Proposition 6.18](#). ■

Remark 6.3. For a linear map $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$, measurability of E implies the measurability of $T(E)$ because we are working in the same dimension. If $T : \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$ for $d' < d$, then measurability of E does not always imply the measurability of $T(E)$. For instance, suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^1$ be the projection onto the first component.

Consider $E = A \times \{0\}$, where A is the non-measurable set constructed in [Theorem 3.11](#). Then E has outer measure 0 as it can be covered using the 0-measure rectangle $[0, 1] \times [0, 0]$. So $E \subseteq \mathbb{R}^2$ is measurable. But $T(E) = A \subset \mathbb{R}$ is **NOT** measurable.